

PCR-6

Physical Nature and Minimal Identity of the Neutrino Carrier

Terminal carrier identity, standing spectral sectors, and branch-stable physical nature after joint neutrino closure

Scope of the result

The theorem applies to the two source-qualified vectorlike-joint sectors completed by PCR-5E: the D3-1.3/public-summary bounded-accidental sector and the D3-1.3/public-summary exact-global sector. Every accepted realization in either sector contains one terminal continuation-bearing neutrino carrier. Its standing spectral core consists of three nondegenerate light sectors and one rank-three degenerate heavy sector. The carrier core is common to both sectors, while the complete minimal identity packet retains target-dependent symmetry and conjugation data. The continuum PCR-5E quotient and all unresolved parent sectors remain unchanged.

Abstract

PCR-5E establishes two exact simultaneous neutrino sectors within a source-qualified vectorlike-joint comparison class: a D3-1.3/public-summary sector with bounded accidental $B - L$, and a D3-1.3/public-summary sector with exact global $B - L$. Both sectors contain continuum many SCIC-distinct joint realizations. The present paper addresses a different question: within one accepted realization, which target-visible structures are internal standing sectors, and which structures constitute independently complete terminal carriers?

Internal sectors are defined by target-relative physical use rather than by eigenvector labels, field names, or a chosen matrix presentation. Their source-first reconstruction gives an activity-reduced groupoid natural over each completed moduli groupoid. For the vectorlike-joint source, the exact mass map

$$\mathbb{M}_{m_0, \lambda, s_E} = \mathcal{U}_L(s_E) \begin{pmatrix} \Sigma(m_0) & 0 \\ 0 & H_\lambda \mathbf{1}_3 \end{pmatrix} \mathcal{U}_R(s_E)^\dagger$$

has three nondegenerate light spectral projectors and a threefold-degenerate heavy eigenspace. The light projectors remain standing-distinct through the retained oscillation and charged-current boundary traces. Every target-admitted witness on the heavy subspace is equivariant under the lawful $U(3)_H$ action and therefore descends to a single rank-three projector. The exact standing spectral core is consequently composed of three light sectors and one heavy sector.

Terminal carrier identity is evaluated only after the complete terminal witness and continuation interface has been activated. No individual light sector, proper light aggregate, heavy sector, proper light-heavy aggregate, or irreducible relation is terminal-complete. Eligibility for the terminal bearer role is formed independently of parent SCIC identity. A separate source-support exhaustion theorem proves that every independently eligible candidate capable of entering a complete one-apex assignment must contain the full deletion-essential support of the parent realization and is therefore lawfully equivalent to the source-connected parent carrier. Standing Carrier Identity Closure is applied only after this source and physical uniqueness result, and a fresh within-realization Role-Occupancy calculation yields one live bearer class and one compatible assignment orbit.

Thus every accepted realization X in either completed sector satisfies

$$|\text{TermCar}(X)| = 1, \quad |\pi_0 \text{SpecCore}(X)| = 4.$$

The terminal object is a local hybrid neutral-fermion carrier with a vectorlike Dirac mass-map core, common charged-current boundary incidence, and target-essential matching and continuation structure. Its carrier core is natural across both completed moduli and common to the accidental and exact-global sectors. Their full minimal identity packets nevertheless remain distinct because accidental LNV permissibility and exact-global protection are standing target data.

This one-carrier result does not select a unique PCR-5E realization. The continuum joint quotient and its cross-realization over-residue diagnosis remain unchanged, as do the unresolved PCR-5E parent sectors. Metric masses, pole structures, Higgs/Yukawa representations, and mixing coordinates remain downstream representations of already-standing carrier and sector structure; they do not generate terminal identity retroactively.

Keywords. AASC; neutrino carrier; standing sector; terminal identity; vectorlike Dirac source; spectral stratification; SCIC; Role-Occupancy Closure; target extension; no-backflow.

Principal theorem

Branch-stable one-carrier closure on the completed vectorlike-joint sectors

Let

$$\mathcal{B}_6^{\text{ready}} = \{b_{\text{acc}}, b_{\text{glob}}\}$$

be the two PCR-5E v1.4.0 target-and-comparison-qualified vectorlike-joint child scopes. For every $b \in \mathcal{B}_6^{\text{ready}}$ and every accepted one-apex realization $X \in \text{PhysMod}_{E, \text{VJ}}^b$, there exist:

1. a source-first activity-reduced internal-sector groupoid $\text{Sect}_{\text{act}}^b(X)$;

2. a carrier-bearing spectral core

$$\text{SpecCore}^b(X) = \{S_1, S_2, S_3, S_H\}$$

consisting of three nondegenerate light rank-one sectors and one heavy rank-three degenerate sector;

3. a source-faithful terminal parent bearer $C_\nu^{\text{VJ}}(X)$ whose complete terminal continuation signature is defined and target-complete;
4. a branch-indexed minimal identity packet

$$\text{Identity}_{\min}^b(X) = \text{NatureCore}(X) \oplus \text{StandingDecoration}^b(X).$$

No proper sector, proper aggregate, or irreducible relation has a complete terminal signature. Eligibility for the target-derived bearer role is independent of parent identity. Every independently eligible candidate that is also terminal-role-complete and one-apex compatible is proved, by source/deletion exhaustion, to contain the full parent support and to be lawfully equivalent to $C_\nu^{\text{VJ}}(X)$. The live within-realization terminal bearer residue is

$$\text{Residue}_{\text{term}}^{\text{live}}(X) = \{[C_\nu^{\text{VJ}}(X)]\},$$

and the terminal one-role assignment quotient is singleton:

$$\left| \Sigma_{C_X}(T_{\text{term}}^{\nu,b}) / \Gamma_{C_X,T}^{\text{lawful}} \right| = 1.$$

Consequently

$$\text{RealizationLicensed}(C_X, T_{\text{term}}^{\nu,b}), \quad |\text{TermCar}_b(X)| = 1.$$

The bearer kind is a local hybrid neutral-fermion carrier with a vectorlike Dirac mass-map core. Its carrier core and 3 + 1 sector topology are natural across the completed moduli and common to both children; the full minimal child packet differs only by target-dependent symmetry/conjugation decoration. The PCR-5E continuum joint quotient, cross-realization over-residue diagnosis, and unresolved parent branches remain unchanged.

Scope and interpretation

Established here

The paper establishes the branch-indexed internal-sector construction, exact 3 + 1 carrier-bearing spectral stratification, moduli naturality, terminal activation, sector/aggregate/parent/relation signature comparison, spectral-sector nonpromotion, source-faithful one-bearer realization, kind classification, within-realization Role-Occupancy Closure, layered minimal identity packet, cross-child carrier-core comparison, and metric/Higgs/Yukawa no-backflow theorem.

Not established here

The paper does not complete the PCR-5E parent joint problem, select one joint realization class, identify the unresolved D3-1.1 or complete-D4 branches, generalize the result to every neutrino carrier architecture, or prove a universal one-carrier theorem for charged leptons and quarks. It also does not identify four terminal particles: the 3 + 1 result concerns standing sectors of one target-qualified carrier core.

The result is exact relative to the two PCR-5E v1.4.0 child target fingerprints, the corresponding source-qualified vectorlike-joint comparison classes, the finite source-presentation discipline inherited from PCR-3 and PCR-5B, and the fixed-domain SCIC and Role-Occupancy framework. Supplementary formal data document provenance and reproducibility but do not replace the semantic proofs.

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1 Introduction

1.1 Carrier identity after simultaneous target closure

The neutrino carrier programme separates several questions that conventional field-theoretic notation often presents together. The oscillation target first determines the physical work that must be performed without naming its occupant. Source-first analysis then classifies admissible carrier architectures, simultaneous physicality eliminates incompatible constructions, and Standing Carrier Identity Closure determines which surviving presentations represent the same target-relative carrier. PCR-5C and PCR-5D1–D5 extend the oscillation target by direct kinematics, lepton-number-violating response, active-sector closure, cosmological transport, and symmetry or charge structure. PCR-5E closes two simultaneous vectorlike-joint sectors within that enriched setting [Mal26ai; Mal26d; Mal26ad; Mal26ac; Mal26c; Mal26af; Mal26i; Mal26l; Mal26g; Mal26p; Mal26m; Mal26ag; Mal26f; Mal26ah; Mal26o].

PCR-5E classifies joint realizations across a comparison category. It does not determine how many independently complete carriers occur inside any one accepted realization. PCR-6 addresses that within-realization problem:

Which target-visible substructures are standing internal sectors, which of them are independently complete terminal bearers, and what minimal physical nature carries the full simultaneous neutrino target?

This distinction is essential. Continuum many SCIC-distinct joint realizations can each contain one carrier, while one realization can in principle contain several carriers. Neither multiplicity determines the other.

1.2 Completed simultaneous sectors

The theorem is defined on the two completed PCR-5E vectorlike-joint sectors

$$\mathcal{B}_6^{\text{ready}} = \{b_{\text{acc}}, b_{\text{glob}}\},$$

with

$$\begin{aligned} b_{\text{acc}} &= (\text{THUCL-D3-1.3}, T_{D4}^{\text{pub95}}, T_{\text{acc}}^{(4)}, C_{\text{VJ}}^{\text{acc}}), \\ b_{\text{glob}} &= (\text{THUCL-D3-1.3}, T_{D4}^{\text{pub95}}, T_{\text{glob}}, C_{\text{VJ}}^{\text{glob}}). \end{aligned}$$

For each branch b , PCR-5E supplies a target-and-comparison-qualified physical groupoid $\text{PhysMod}_{E, \text{VJ}}^b$, a joint quotient $Q_{E, \text{VJ}}^b$, the complete continuation structure, and the exact unresolved complement. D3-1.1, the complete D4 cosmological target, the non-RCF-finite D3-1.3 remainder, and five D5 comparison classes remain outside the completed domain.

1.3 Role, realization, sector, and bearer multiplicities

Four cardinalities remain logically distinct:

1. the number of terminal roles forced by the target;
2. the number or moduli type of joint realization classes across the PCR-5E child category;
3. the number of standing internal sectors inside one accepted realization;
4. the number of independently complete terminal carriers inside that realization.

Both completed sectors have one terminal carrier role, while their joint realization quotients contain at least continuum many classes. The displayed vectorlike mass matrix has six algebraic eigenvectors. None of these facts determines the within-realization carrier count.

1.4 Spectral standing in the vectorlike-joint realization

The intrinsic vectorlike mass map is

$$\mathbb{M}_{m_0, \lambda, s_E} = \mathcal{U}_L(s_E) \begin{pmatrix} \Sigma(m_0) & 0 \\ 0 & H_\lambda \mathbf{1}_3 \end{pmatrix} \mathcal{U}_R(s_E)^\dagger.$$

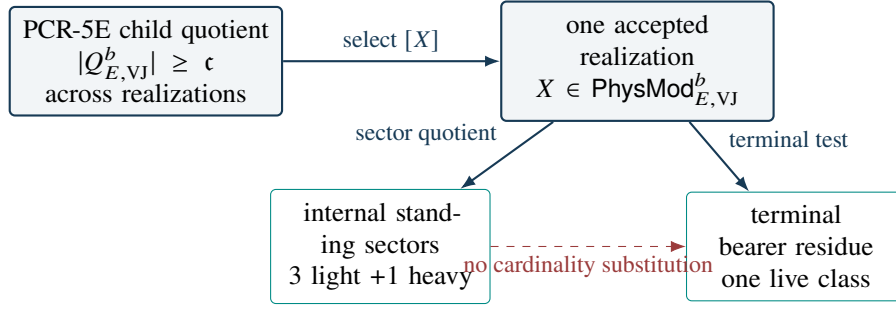


Figure 1: Carrier multiplicity at two levels. PCR-5E counts joint realization classes across the completed child groupoid; PCR-6 counts independently complete terminal carriers inside one accepted realization. Internal sector cardinality is intermediate and carries no automatic bearer interpretation.

Its three light poles are nondegenerate and remain distinguishable through the relative propagation and charged-current boundary traces. The heavy block is exactly degenerate and admits a lawful $U(3)_H$ action. The central spectral question is therefore not how many eigenvectors can be labelled, but which distinctions survive the complete target-relative witness and continuation structure.

1.5 Main results

The analysis establishes four principal conclusions.

First, the activity-reduced internal-sector quotient contains exactly three nondegenerate light standing sectors and one rank-three heavy standing sector. The heavy basis labels collapse because every activated heavy-subspace witness descends through the lawful $U(3)_H$ orbit, while deletion of the aggregate heavy projector changes the D3/D5 target trace.

Second, no proper sector, proper aggregate, or irreducible relation is terminal-complete. The light aggregate lacks the heavy response and matching structure; the heavy sector lacks the oscillation, direct-kinematic, and cosmological light-sector response; and relation objects lack independent persistence and boundary reconstruction.

Third, terminal eligibility is formed independently of parent SCIC identity. Source-support exhaustion then proves that any eligible, terminally complete, one-apex compatible candidate must contain the full deletion-essential support of the parent and is lawfully equivalent to the source-connected carrier. A fresh Role-Occupancy calculation therefore yields one terminal carrier inside every accepted realization.

Fourth, the carrier core is natural across both completed moduli and common to the accidental and exact-global sectors. Their complete minimal identity packets remain target-distinct because accidental LNV permissibility and exact-global protection are standing data.

1.6 Organization

Sections 2–4 fix the completed target scopes and the physical-nature problem. Sections 5–8 construct the internal-sector quotient and establish its moduli naturality. Sections 9–12 activate the terminal interface and compare the signatures of sectors, aggregates, relations, and the parent. Sections 13–18 establish bearer existence, exhaust alternatives, classify physical kind and quantum numbers, and apply within-realization Role-Occupancy Closure. Sections 19–22 determine the minimal identity packet, compare the two completed sectors, prove the projection boundary, and state the main theorem. The final sections record consequences, counterexample burdens, and conditions for extending the result.

Nonclaim 1.1 (No parent or universal promotion). The theorem does not say that every possible neutrino realization has the vectorlike-joint physical nature proved here. It does not complete the PCR-5E parent residue, and it does not license transport to charged leptons or quarks. Those claims require new target and source theorems.

2 Foundational setting and target separation

2.1 Target identity and dependency discipline

Theorem 2.1 (P6-00 – import and no-backflow lock). *There is a typed import graph for PCR-1 through PCR-5E in which every imported object retains its target, comparison class, continuation scope, and readiness status. In particular:*

1. *D3-1.1 and D3-1.3 remain distinct;*
2. *D4 public-summary and D4 complete remain distinct;*
3. *PCR-5E child closure does not complete any parent branch;*
4. *no mass, Yukawa, PMNS, metric, likelihood, or physical-nature datum enters the PCR-6 target before terminal activation;*
5. *every theorem below is indexed by $b \in \mathcal{B}_6^{\text{ready}}$.*

Proof. PCR-5E v1.4.0 and its ARC update provide the branch authority and exact child fingerprints. The D3 programme-separation theorem blocks backward transport from 1.3 to 1.1, and the D4 theorem preserves the complete-likelihood gate. The PCR-5E E-H records identify the two completed child certificates and declare every parent branch NotReady or ROOS at one or more mandatory fields. We form the PCR-6 import graph using only those typed edges. Any edge dropping a target tag or using a parent-incomplete field is rejected fail-closed. The premetric no-backflow clauses follow from the fixed-domain, MGN, Higgs, and SCIC imports [Mal26a; Mal26ae; Mal26q; Mal26s; Mal26aj; Mal26h; Mal26ak]. $\square$

2.2 Roles of the inherited results

The inherited results occupy distinct logical positions. Their roles are summarized below; none may be substituted for another without changing the proof.

Source	Imported object	Use in PCR-6
PCR-1/PCR-2	Carrier-neutral role and complete target-facing interface	Defines boundary, response, compatibility, and semantic target uses without naming a carrier.
PCR-3/PCR-4	Source-first carrier grammar and one-apex physicality	Supplies the source-completeness and simultaneous-realization discipline used at the within-realization level.
PCR-5/5A/5B	One-role ROC, seven-channel packet, exact SCIC atlas	Separates source factorization, standing carrier identity, and terminal role occupancy.
PCR-5C	Restriction, fibres, no merger, joint target, source composition	Supplies the functorial and groupoid structure used to compare the completed moduli.
D1–D5	Target-tagged standing coordinates and failure boundaries	Supplies absolute scale, LNV, closure, population, and symmetry data without promoting any one coordinate to carrier identity.
PCR-5E	Two completed E-H1–E-H10 data sets	Fixes the two domains on which PCR-6 is proved and preserves the unresolved parent boundary.
SCIC/ROC/fixed-domain	Complete continuation signatures and unique assignment-orbit criterion	Supplies the terminal identity and realization closure principles.
MGN5/MGN6/Higgs arc	Matrix, spectrum, mixing, Higgs, and Yukawa no-promotion	Blocks downstream representations from generating premetric carrier identity.

Source	Imported object	Use in PCR-6
SM carrier arc	Matter-carrier and generation nonpromotion discipline	Calibrates the difference between carrier inventory, flavor transition, and generation governance.

2.3 Boundary of the completed sectors

Theorem 2.2 (P6-R1 – ready-scope family and parent nonpromotion). *Let $\mathcal{P}_E$ be the full PCR-5E parent target family and let $\mathcal{B}_6^{\text{ready}} \subset \mathcal{P}_E$ be the two completed vectorlike-joint children. Physical-nature analysis defines a functor only on the full subgroupoid over $\mathcal{B}_6^{\text{ready}}$. No result of PCR-6 induces a selector, completion map, or exclusion map on $\mathcal{P}_E \setminus \mathcal{B}_6^{\text{ready}}$.*

Proof. The unresolved parent branches lack one or more of the final quotient, continuation, source-sufficiency, failure-frontier, or finality components required by E-H4 and E-H7–E-H10. A physical-nature classifier presupposes an accepted object and cannot construct the missing accepted domain. Therefore its domain is the ready child groupoid only. Extending it to the parent would require a new target/source certificate and is a TargetUpdate, not a consequence of the child theorem. $\square$

Corollary 2.3 (No physical-nature repair of parent readiness). *Even if every $X \in \text{PhysMod}_{E,\text{VJ}}^b$ has one terminal bearer, PCR-6 cannot infer a singleton or complete residue for the PCR-5E parent theory.*

3 Completed vectorlike-joint sectors

3.1 Inherited simultaneous structure

For each completed sector $b \in \mathcal{B}_6^{\text{ready}}$, write

$$\mathcal{H}_E^b = (EH1^b, \dots, EH10^b).$$

These data comprise the exact joint target, source category, one-apex physical category, child quotient, restriction maps, standing-coordinate incidence, continuation monoid, source-sufficiency profile, failure frontier, and declared scope.

Theorem 3.1 (P6-01 – branch-indexed PCR-5E finality). *The packages $\mathcal{H}_E^{b_{\text{acc}}}$ and $\mathcal{H}_E^{b_{\text{glob}}}$ are complete for within-realization sector and terminal-bearer analysis. The parent package is not complete and carries the exact status $\text{NotReady}_{\text{PCR5E}}$.*

Proof. The two child packages contain all ten E-H fields and the E-S1–E-S7, E-S4A, and E-S7A certificates. In particular, they contain a semantic-use saturation theorem, source-qualified comparison category, one-apex physical groupoid, exact child quotient, complete fixed-base foliation, standing-coordinate relations, source-deletion profile, and finality declaration. Every terminal use introduced below factors through one of those fields together with the independently formed physical-nature target. The parent family lacks the complete final quotient and terminal continuation/finality data on at least one branch, so the same factorization is unavailable there. $\square$

3.2 Parameterization of the completed sectors

The exact-global child has objects

$$X_{\text{glob}}(m_0, \lambda, \tau), \quad (m_0, \lambda, \tau) \in I_m^\circ \times I_E \times I_\tau^\circ,$$

with the D2 source and amplitude fixed at zero. The bounded-accidental child has objects

$$X_{\text{acc}}(m_0, \lambda, \tau, z), \quad (m_0, \lambda, \tau, z) \in I_m^\circ \times I_E \times I_\tau^\circ \times I_{76},$$

with an accepted higher-dimensional LNV response. The parameter $s = s_E > 0$ is fixed by the D3/D5 strict-margin product-box theorem.

The intrinsic vectorlike source depends on (m_0, λ, s_E) . The parameter τ belongs to the independently authorized common D1/D4 extension and has an exact reduct to the same PCR-5B base class. On the accidental branch, z is an additional response coordinate and does not alter the intrinsic vectorlike mass-map core.

3.3 Across-realization and within-realization multiplicity

PCR-5E has already established

$$|Q_{E,VJ}^b| \geq \mathfrak{c}, \quad \text{Status}_{\text{ROC}}^E(b) = \text{OverResidue}$$

for both completed sectors. Thus each one-role child target contains continuum many live realization classes across its comparison category. PCR-6 instead fixes one object X and asks whether its internal subobjects generate one or several complete terminal bearers.

Criterion 3.2 (Two-level ROC firewall). A within-realization conclusion $|\text{TermCar}_b(X)| = 1$ may not be substituted for $|Q_{E,VJ}^b| = 1$, and the cross-realization inequality $|Q_{E,VJ}^b| \geq \mathfrak{c}$ may not be substituted for $|\text{TermCar}_b(X)| > 1$.

4 The target-relative physical-nature problem

4.1 Physical-nature target

For each $b \in \mathcal{B}_6^{\text{ready}}$, the physical-nature target $T_{\text{nat}}^{\nu,b}$ asks for the terminal bearer structure of an already accepted object $X \in \text{PhysMod}_{E,VJ}^b$. It contains the following roles:

1. preserve the complete child target and its one-apex physical identity;
2. expose every internal subobject or relation with target-relevant standing;
3. determine which such subobjects are independently complete under all licensed terminal uses;
4. classify the physical kind of every complete bearer;
5. derive the terminal role set, eligibility, compatibility, and lawful automorphisms before assignment;
6. construct the minimal target-relative identity packet and its downstream projection boundary.

Theorem 4.1 (P6-02 – target-indexed physical-nature target formation). *For each $b \in \mathcal{B}_6^{\text{ready}}$, there is a target certificate*

$$T_{\text{nat}}^{\nu,b} = (D_b, R_b, W_b, U_b, \sim_b, \partial\text{Fail}_b, \text{Update}_b)$$

formed before sector inspection and independent of any carrier count, spectral rank, source-family name, Dirac/Majorana label, or point-locality assumption.

Proof. The domain D_b , inherited target roles, source comparison category, boundary interfaces, joint standing coordinates, continuation monoid, and failure/update boundary are supplied by the E-H package. The new target role is the within-realization bearer question. Its semantic uses are generated by deleting, continuing, comparing, and reusing subobjects while preserving the accepted child target. These operations can be stated without naming a candidate bearer. Removing all matrix sizes, eigenvalue labels, field-family names, and desired cardinalities leaves the target unchanged, proving target neutrality. $\square$

Proposition 4.2 (Target-derived terminal role and internal-role firewall). *For each ready child, the target-derived terminal role set is*

$$R_{\text{term}}^{\nu,b} = \{r_{\text{bear}}^b\}.$$

The role is a target-derived bearer-capability role. It requires a same-child persistent bearer capable of the complete boundary, propagation, continuation, matching, D1–D5 role interfaces, and one-apex physical use appropriate to branch b ; it does not contain the numerical or categorical continuation signature of any particular realization. This one-role statement does not identify internal sectors with terminal roles and does not forbid several sectors, relations, or source blocks inside one bearer. A proper component enters the live terminal residue only after independently formed eligibility, terminal-role completeness, and assignment compatibility are established.

Proof. PCR-5E forms one complete-carrier role for each source-qualified joint child before its cross-realization residue is evaluated. D1–D5 add standing coordinates, processes, histories, and symmetry conditions to that role but do not introduce separately forced terminal occupants. The new PCR-6 target asks whether a subobject can itself occupy the bearer role under the complete child use. Consequently the role set remains one at terminal resolution, while node, aggregate, and relation loci remain internal descriptive objects until promotion is proved. Eligibility for the role is formed later from target typing, source provenance, stable boundary/role traces, and capability; it is not defined by equality with the parent identity signature. This prevents both directions of cardinality substitution: one role does not imply one sector or one carrier class, and several sectors do not imply several terminal roles. $\square$

4.2 Common carrier structure and branch-specific data

The two targets share the same vectorlike source category, boundary, light/heavy mass-map core, D3 active-closure packet, D4 public-summary environment, matching structure, and terminal one-role architecture. They differ in their D2/D5 relation:

- b_{acc} : bounded accidental $B - L$ through $d \leq 4$, with an accepted optional higher-dimensional LNV response;
- b_{glob} : exact global $B - L$ with a continuation-stable protector and a defined zero D2 subfibre.

This difference is retained throughout and is not treated as a presentation change.

Nonclaim 4.3 (No cross-child equality in target formation). The two physical-nature targets are not identified before the branchwise terminal and minimal-identity calculations. Cross-child comparison is a later theorem.

5 Target-relative internal sectors

5.1 Sector uses before sector presentations

A sector is not defined as an eigenvector, a named field, a block of a displayed matrix, or a source-family label. It is a subobject or irreducible relation of an accepted realization whose retention, deletion, continuation, boundary incidence, or compatibility can change a target-authorized internal use while the parent target remains fixed.

Let $\text{Sub}(X)$ denote the finite-presentation subobject and finite-relation category generated by the activity-reduced source trace of X . Its objects include carrier-support subgraphs, spectral projectors or measures, boundary-incidence objects, matching subdiagrams, population components, symmetry actions, conjugation/LNV components, and irreducible finite higher relations.

Definition 5.1 (Semantic sector use). A semantic internal-sector use of $A \in \text{Sub}(X)$ is a target-compatible map that evaluates at least one of:

Defined, Boundary, Persist, Continue, Delete, Compat, Fail, RelInc, Reuse

on A or on its incidence with the parent, and is invariant under lawful child-source redescription.

Definition 5.2 (Internal-sector trace). Let $\mathcal{U}_{\text{sec}}^b$ be the internal continuation category generated in Section 9. The complete semantic sector trace is

$$\tau_{\text{sec}}^{b,X}(A) = (u(A), w(u(A)), \text{Inc}_X(u(A)))_{u,w},$$

where $u \in \mathcal{U}_{\text{sec}}^b$ and w ranges over target-authorized internal-sector uses.

Theorem 5.3 (P6-03 – semantic internal-sector class). For every $b \in \mathcal{B}_6^{\text{ready}}$ and $X \in \text{PhysMod}_{E,VJ}^b$, equality of complete semantic sector traces defines an extensional equivalence relation on $\text{Sub}(X)$. The quotient

$$\text{Sect}_{\text{sem}}^b(X) = \text{Sub}(X) / \equiv_{\tau_{\text{sec}}}$$

contains every target-relevant internal node, aggregate, and irreducible relation while excluding labels and source distinctions that change no internal use.

Proof. Trace equality is reflexive, symmetric, transitive, and stable under composition because the continuation family is closed. If two subobjects differ only by a lawful basis, gauge, source-isomorphism, chart, or degenerate-eigenspace representative, covariance of every semantic use gives equal traces. Conversely, if a subobject difference

changes target-relative definedness, boundary incidence, continuation, deletion response, compatibility, failure, or reuse, that difference appears in the trace by construction. Thus the quotient is neither finer than target use nor coarser than a target-visible internal distinction. $\square$

5.2 Node, aggregate, relation, and parent levels

The semantic class distinguishes four support modes:

1. *node sectors*: connected source-support or spectral components;
2. *aggregate sectors*: finite unions of node sectors with induced incidence;
3. *relation sectors*: irreducible finite relations whose deletion changes standing although no new node is added;
4. *the parent*: the complete source-connected carrier core with all deletion-essential incidence.

This support hierarchy is descriptive at the internal level. None of its objects is terminal until the activation and promotion theorems are applied.

Remark 5.4 (Why population and symmetry data enter the semantic class). The D4 population history and D5 action/protector can be internal standing relations even when they are not carrier nodes. Their deletion changes the complete child target, so they must remain represented in $\text{Sect}_{\text{sem}}^b(X)$; the later terminal analysis determines their bearer status.

5.3 Physical-nature semantic-use descent

The physical-nature target adds a new comparison question but no hidden source primitive. Its semantic uses must therefore descend through the complete carrier-semantic trace already supplied by PCR-3 together with the target/source incidence trace closed by PCR-5E.

For $A \in \text{Sub}(X)$, define the combined relative trace

$$\Theta_{\text{nat}}^{b,X}(A) = (\chi_{\text{car}}^{(3)}(A), \tau_E^b(A), \tau_{\text{sec}}^{b,X}(A)),$$

where $\chi_{\text{car}}^{(3)}$ is the activity-reduced PCR-3 carrier trace and τ_E^b is the complete target/source incidence trace imported through E-H1–E-H9.

Theorem 5.5 (P6-S0 – physical-nature semantic-use descent and relative source completeness). *For every ready child b , every accepted realization $X \in \text{PhysMod}_{E,\text{VJ}}^b$, and every same-child semantic use of a candidate internal bearer, sector, aggregate, relation, or parent object, that use factors through $\Theta_{\text{nat}}^{b,X}$ up to lawful source and presentation equivalence. Consequently the source theory used in P6-04 is complete relative to the E-H2 child comparison category. A proposed use that does not factor through this trace is either a target/source update or lies outside the completed child scope; it is not an omitted same-scope sector.*

Proof. PCR-3 proves that the activity-reduced carrier trace records every same-target carrier-semantic difference in support, persistence, derivation status, constituent structure, intrinsic typed data, target attachment, and physical incidence. PCR-5E E-S1/E-S2 adds the complete simultaneous target/source incidence, matching, background, continuation, higher-arity, and common-cone trace for the two E-H2 child categories. The PCR-6 physical-nature target is formed from these already frozen objects and adds only target-relative bearer uses: locality or support type, persistence and reuse, constituent retention, boundary reconstruction, deletion, compatibility, failure, and lawful redescription. It adds no new source constructor or endpoint-dependent carrier label.

Let f be a same-child semantic use. If changing its argument changes target typing or the source domain, the change is an anchor and therefore a declared update. With anchors fixed, any standing-bearing change is tensor content. The PCR-3 carrier trace records the carrier-facing tensor data, the PCR-5E trace records the simultaneous joint tensor data, and $\tau_{\text{sec}}^{b,X}$ records their restriction to the candidate subobject and its incidence with X . Hence f factors through $\Theta_{\text{nat}}^{b,X}$. A residual change preserving all three trace factors is skin or provenance and cannot define a new standing sector by the fixed-domain no-second-classifier theorem. This proves both semantic-use descent and relative source completeness. $\square$

6 Source representation and activity reduction

6.1 Within-realization source theory

For each completed sector, define a conservative within-realization source theory $S_{\text{sec}}^{v,b}$ over the exact E-H2 source category. Its sorts are:

BearerSupport, LightSpectral, HeavySpectral, BoundaryInc, Propagation, Matching,
Conjugation, LNVResponse, ActiveClosure, Population, Background, SymmetryAction,
Protector, Constituent, Collective, HigherRelation, CommonApex.

The constructors are restrictions of the PCR-3/PCR-5E source operations: Retain, Attach, Inflow, Fuse, CoCarry, Project, Boundary, Constituent, Collective, Match, Continue, and Amalgamate.

Theorem 6.1 (P6-04 – source-first sector grammar). *Every finitely presented semantic sector architecture in $\text{Sect}_{\text{sem}}^b(X)$ has a source-coherent normal form in $S_{\text{sec}}^{v,b}$, and every generated normal form has a semantic interpretation as a subobject or irreducible relation of X . The construction is indexed before sector cardinality or terminal disposition is inspected.*

Proof. By Theorem 5.5, every same-child physical-nature use factors through the complete PCR-3/PCR-5E carrier and incidence trace. The accepted E-H2 source category is a finite source-incidence theory inherited from PCR-3 and saturated in PCR-5E by E-S1/E-S2. Restricting that grammar to subobjects of one source point therefore yields every node, aggregate, and finite-relation presentation admitted by the child comparison class. Each semantic subobject has finite carrier support, typed incidence, and a source reduct; source-first normalization expresses it using the listed sorts and constructors. Conversely, every coherent generated term is a typed subdiagram of the accepted source and therefore has a semantic use. Since the grammar is defined from E-H2 before the target endpoint is evaluated, no one-carrier backfitting occurs. $\square$

6.2 Activity reduction

The vectorlike-joint source contains both active and declared-null target-extension cells. Activity is determined by source equations and target attachment, not by numerical magnitude. In particular, the vectorlike-heavy D4 population kernels are source-false even though the old D3/D5 vectorlike source remains active.

Definition 6.2 (Sector activity). For a sector-source incidence e at source point x , define

$$\text{Act}_x(e) = 1$$

iff its source constitutive equation is nonzero or its target attachment is enabled at x . The value is 0 when the source equation fixes the incidence null and $\perp$ when the source theory does not define the required incidence.

Theorem 6.3 (P6-05 – sector activity reduction). *Pointwise activity reduction removes every source-null incidence and disconnected latent sector, retains every nonnull participating incidence without a weak-coupling threshold, and preserves every deletion-essential higher relation. It is congruent under lawful source isomorphism and yields an activity-reduced sector groupoid $\text{Sect}_{\text{act}}^b(X)$.*

Proof. The PCR-3/PCR-5E activity semantics are pointwise and typed. Desugaring support references into explicit incidence makes adjacency and target-path closure functorial. A nonzero source incidence remains active regardless of magnitude; a source-null incidence is removed before path closure; and a higher relation is retained whenever its deletion changes a semantic sector trace. Lawful source isomorphisms preserve constitutive equations and target attachment, so activity reduction is congruent. $\square$

6.3 Activity dispositions in the two sectors

On both children the intrinsic light/heavy vectorlike source, common charged-current boundary, active closure, threshold/matching, light D4 populations, and D5 action block are active. All vectorlike-heavy D4 population incidences are source-null. On the exact-global child the LNV source is absent and the D2 zero cell is a response value, not an active source sector. On the accidental child an LNV relation sector is active when $z \neq 0$ and is removed at the source-null zero point.

Corollary 6.4 (Accidental activity stratification). *The accidental sector functor has two activity strata:*

$$\mathcal{Z}_0 = \{z = 0\}, \quad \mathcal{Z}_\times = I_{76} \setminus \{0\}.$$

The carrier-bearing source core is identical on both strata; only the optional LNV relation sector changes activity.

7 Exact carrier-bearing spectral stratification

7.1 Intrinsic vectorlike mass map

At fixed (m_0, λ, s_E) , the intrinsic vectorlike source has the exact mass map

$$\mathbb{M} = \mathcal{U}_L(s_E) \begin{pmatrix} \Sigma(m_0) & 0 \\ 0 & H_\lambda \mathbf{1}_3 \end{pmatrix} \mathcal{U}_R(s_E)^\dagger, \quad \Sigma(m_0) = \text{diag}(m_1, m_2, m_3).$$

The light masses are positive and pairwise distinct because the target fixes nonzero mass-squared differences on I_m° . The heavy threshold is separated from the light spectrum and is exactly triply degenerate.

Define the source-covariant projectors

$$P_i = \mathcal{U}_L \begin{pmatrix} E_{ii} & 0 \\ 0 & 0 \end{pmatrix} \mathcal{U}_L^\dagger, \quad i = 1, 2, 3,$$

$$P_H = \mathcal{U}_L \begin{pmatrix} 0 & 0 \\ 0 & \mathbf{1}_3 \end{pmatrix} \mathcal{U}_L^\dagger.$$

The individual heavy rank-one projectors depend on a basis inside the degenerate heavy subspace; P_H does not.

7.2 Light-sector standing

Lemma 7.1 (Light-projector distinction). *The three light projectors define three pairwise distinct internal standing sectors.*

Proof. Suppose two light projectors had equal complete internal traces. Then all licensed propagation and boundary uses would agree. But the retained propagation phase functions contain distinct nonzero mass-squared differences, and the charged-current boundary incidence records the corresponding columns of the boundary-to-propagation map modulo only lawful rephasing and permutation. Identifying two projectors would either remove a target-visible phase frequency, change the no-repair endpoint/population weights, or require a degenerate spectral relation excluded by the fixed light interval. Hence the traces differ. $\square$

7.3 Heavy degeneracy quotient

Let $\Gamma_H = U(3)$ act on the heavy degenerate eigenspace. Under

$$N_L \mapsto WN_L, \quad N_R \mapsto WN_R, \quad W \in U(3),$$

the heavy mass term $H_\lambda \mathbf{1}_3$, exact vectorlike charge packet, anomaly data, threshold, and aggregate heavy charged-current response remain unchanged after the compensating lawful source/basis action.

Lemma 7.2 (Heavy-subspace collapse). *Every individual heavy basis projector has the same complete internal-sector trace up to the lawful Γ_H action. The heavy standing quotient therefore has one object, represented by the rank-three projector P_H .*

Proof. Let the heavy charged-current block be $K_H = -s_E U$. Under a lawful heavy-basis change $W \in U(3)$ it becomes $K_H W^\dagger$, while the heavy mass and charge operators remain $H_\lambda \mathbf{1}_3$ and $-\mathbf{1}_3$. Every target-authorized heavy response has the covariant form

$$K_H W^\dagger f(H_\lambda^2 \mathbf{1}_3) W K_H^\dagger = K_H f(H_\lambda^2 \mathbf{1}_3) K_H^\dagger,$$

or is a trace, determinant, threshold, anomaly, matching, or continuation functional of the same scalar heavy block. The oriented matching diagram is transported by the same W on its heavy leg, so its physical isomorphism class is unchanged. Thus the complete heavy trace is constant on the $U(3)$ orbit. No target witness distinguishes a basis vector inside the degenerate subspace. Retaining three labeled heavy sectors would preserve an unwitnessed basis distinction and over-resolve the internal standing quotient. The aggregate projector remains standing because deleting it removes the D3 heavy-state response and D5 threshold/matching record. $\square$

Theorem 7.3 (P6-S1 – complete heavy-degenerate witness descent). *Let $E_H = \text{im } P_H$ be the exact heavy eigenspace. For every ready child b , every $X \in \text{PhysMod}_{E, \text{VJ}}^b$, every $W \in U(3)_H$, every target-admitted internal-sector or terminal-use witness w , and every licensed continuation u , the physical witness output satisfies*

$$w(u(W \cdot E_H)) \simeq w(u(E_H)).$$

Equivalently, the complete heavy internal and terminal traces factor through the invariant aggregate projector P_H . The orbit quotient of rank-one heavy presentations is therefore a single standing heavy sector, while deletion of P_H changes the D3/D5 target trace. Hence the exact heavy standing-sector cardinality is one.

Proof. The semantic-use descent theorem [Theorem 5.5](#) reduces the claim to the complete target-authorized carrier, joint-incidence, and sector trace. We verify each activated block.

Propagation. Every heavy propagation or persistence functional depends on the heavy operator through $f(H_\lambda^2 \mathbf{1}_3) = f(H_\lambda^2) \mathbf{1}_3$. It therefore commutes with W , so the propagated subspace and its aggregate projector are unchanged.

Boundary and D3 process response. The heavy charged-current block transforms as $K_H \mapsto K_H W^\dagger$, while an internal heavy leg transforms by W . Every observed channel, Choi block, process-comb contraction, production/decay response, or active-null tester is therefore related by internal conjugation and yields the same boundary map. In particular

$$(K_H W^\dagger)(K_H W^\dagger)^\dagger = K_H K_H^\dagger,$$

so active normalization is invariant.

Charged-lepton flavour. Because the loop kernel is scalar on E_H , the heavy contribution has the invariant form

$$K_H f(H_\lambda^2 \mathbf{1}_3) K_H^\dagger.$$

The exact off-diagonal cancellation follows from unitarity and is unchanged by W . Thus the CLFV value and its typed failure/applicability status descend.

Threshold, matching, and RG/EFT continuation. The physical threshold is the scalar H_λ . Matching maps and oriented threshold diagrams are transported by W on the internal heavy leg, so their physical isomorphism class, rank, orientation, operator-order filtration, continuation domain, and no-backflow status are unchanged.

D5 charge, anomaly, and protection. The heavy vectorlike charge operators and exact-global protector are proportional to the identity on the threefold multiplicity space. All anomaly traces, charge-lattice data, global-action conditions, and protector continuations are invariant under $U(3)_H$.

Applicability, failure, kind, and reuse. Applicability guards and failure frontiers depend on invariant masses, ranks, norms, source equations, response values, and target attachments already shown invariant. The local vectorlike support, constituent status, lawful reuse, and redescription class are likewise unchanged.

These blocks exhaust the physical-nature uses by [Theorem 5.5](#). Hence every complete witness descends through P_H , and all rank-one heavy presentations lie in one lawful orbit. The quotient cannot be coarsened further because deleting P_H removes the heavy-state response, active-closure contribution, threshold, and D5 matching/protector incidence. Thus exactly one heavy standing sector remains. $\square$

7.4 Completeness of the spectral list

Theorem 7.4 (P6-06 – internal sector standing quotient). *For every $b \in \mathcal{B}_6^{\text{ready}}$ and every $X \in \text{PhysMod}_{E, \text{VJ}}^b$, the carrier-bearing spectral subquotient is*

$$\text{SpecCore}^b(X) \simeq \{S_1, S_2, S_3, S_H\},$$

where S_1, S_2, S_3 are the three nondegenerate light rank-one sectors and S_H is the heavy rank-three degenerate sector. Hence

$$|\pi_0 \text{SpecCore}^b(X)| = 4.$$

The complete activity-reduced internal-sector groupoid has the form

$$\text{Sect}_{\text{act}}^b(X) \simeq \int_{S \in \text{SpecCore}^b(X)} \text{RelDecor}^b(X; S),$$

where $\text{RelDecor}^b(X)$ contains boundary, matching, population, symmetry/protector, common-scale, and branch-specific LNV relations.

Proof. The characteristic support of the intrinsic mass map contains exactly three simple light roots and one heavy root of multiplicity three. [Theorem 7.1](#) gives three distinct light standing sectors. The preliminary orbit calculation of [Theorem 7.2](#), strengthened by the complete witness descent theorem [Theorem 7.3](#), proves that every activated internal and terminal witness factors through the aggregate heavy projector and that no heavy basis label survives. The heavy threshold/matching witnesses distinguish P_H from every light projector. No further carrier-bearing spectral root exists. The remaining target-visible internal objects are incidence or response relations attached to this spectral core and are therefore represented in the relation groupoid fibred over the spectral core rather than as additional spectral sectors. $\square$

Theorem 7.5 (P6-07 – sector-cardinality neutrality). *The exact cardinality four is derived from the source spectrum, target-visible phase structure, lawful degenerate-subspace quotient, and activity-reduced internal trace. It is independent of the number of flavor labels, generation roles, thermal modes, displayed matrix columns, or named fields.*

Proof. An ancestry audit of the sector definitions shows that no cardinality-bearing clause contains the numeral three except where the source and target independently supply the three nondegenerate light roots or rank-three degenerate heavy block. Flavor names and matrix labels can be deleted without changing the sector trace. The D4 three-mode compiler supplies population attachments to the already-existing light sectors and is not used to define the carrier-bearing spectrum. Thus the cardinality follows from standing structure rather than presentation backfitting. $\square$

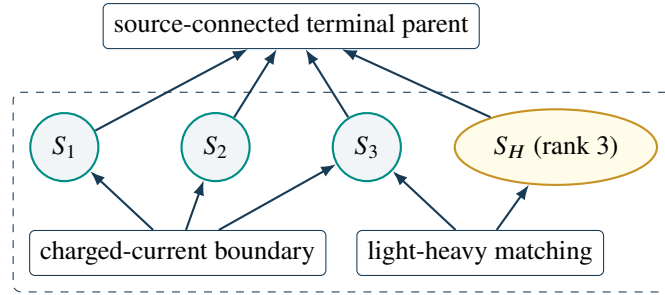


Figure 2: Carrier-bearing sector and relation structure. The three nondegenerate light projectors remain standing-distinct; the degenerate heavy basis collapses to one rank-three sector. Boundary and matching objects are deletion-essential relations, not additional terminal bearers.

Remark 7.6 (Why the result is $3 + 1$, not four terminal particles). The theorem counts standing spectral sectors inside one accepted source-connected parent. Terminal bearer status is a later question. The heavy sector also has internal algebraic multiplicity three, but that multiplicity is not standing at the current target.

8 Naturality across the completed moduli

8.1 Sector functor over the completed groupoids

A pointwise sector theorem would be insufficient because each completed sector is a proper-continuum groupoid rather than a single model. Let Gpd denote the 2-category of small groupoids.

Theorem 8.1 (P6-R2 – moduli-natural sector functor). *For each $b \in \mathcal{B}_6^{\text{ready}}$, activity-reduced sector formation extends to a pseudofunctor*

$$\text{Sect}_{\text{act}}^b : \text{PhysMod}_{E, \text{VJ}}^b \longrightarrow \text{Gpd}.$$

Every lawful morphism of accepted vectorlike-joint realizations induces an equivalence of carrier-bearing spectral cores and a lawful transport of relation decoration. The four-object spectral core is constant on every connected component of $\text{PhysMod}_{E, \text{VJ}}^b$.

Proof. A morphism in $\text{PhysMod}_{E, \text{VJ}}^b$ preserves the source-qualified child fingerprint, one-apex physical trace, target records, lawful source/basis/gauge action, and complete joint standing packet. Restriction of such a morphism to subobjects preserves source activity and semantic sector uses, hence induces a functor on the sector quotient. Composition and identity hold up to the inherited source-presentation coherence, giving a pseudofunctor. The light roots remain pairwise nondegenerate on I_m° , the heavy root remains separated and triply degenerate on I_E , and the fixed $s_E > 0$ preserves the source-connected light/heavy incidence. Therefore no spectral-sector creation, annihilation, or merger occurs inside a connected ready component. $\square$

8.2 Parameter directions

The parameter directions have different standing roles:

- m_0 varies the intrinsic light absolute offset while preserving the three light nondegenerate sectors;
- λ varies the physical heavy threshold while preserving one heavy degenerate sector;
- τ varies the independently authorized common D1/D4 absolute-scale response and preserves all seven PCR-5B base coordinates;
- z varies the accidental D2 LNV response and can change activity of the optional LNV relation sector.

Proposition 8.2 (Carrier-core invariance along m_0, λ, τ). *At fixed child target, the carrier-bearing spectral core and the source-connected parent carrier topology are invariant along all lawful (m_0, λ, τ) paths inside the solved parameter domain.*

Proof. The intervals were frozen away from light degeneracy, heavy/light threshold collision, source-activity, matching, continuation, and failure frontiers. The τ direction is an enriched target extension with an exact base reduct and does not retune the intrinsic vectorlike mass map. Hence the sector topology and parent source incidence remain fixed. $\square$

8.3 Accidental LNV stratification

Theorem 8.3 (Accidental internal-sector stratification). *On the accidental child, the full internal-sector functor has two source-activity strata. For $z \neq 0$, an LNV response relation $\mathbb{V}_z$ is active; at $z = 0$, it is source-null and removed. The carrier-bearing spectral core, parent bearer topology, and terminal disposition are identical on both strata.*

Proof. The D2 source amplitude is linear in z . Its deletion or specialization to zero leaves the light/heavy vectorlike source, boundary, D3, D4, and bounded accidental D5 equations unchanged. PCR-5E source-composition closure shows that the LNV block is optional for target survival. Thus only the relation decoration changes activity; the source-connected carrier core is unaffected. $\square$

8.4 Nature variation types

For a branch b , define

$$\text{NatureVar}_b \in \{\text{Constant}, \text{ConstructiblyStratified}, \text{GenuinelyVariable}, \text{NotReady}\}.$$

The carrier-bearing sector core is constant. After the terminal bearer and minimal identity are established, P6-R2b below upgrades this sector transport to a moduli-natural physical-nature packet. The complete internal decoration is constant on the exact-global child and constructibly stratified on the accidental child by z -activity.

9 Boundary incidence and coherent continuations

9.1 Boundary-to-sector incidence

The charged-current source and detection boundary is operationally fixed before propagation. Let F_α denote the target boundary faces and let P_1, P_2, P_3, P_H denote the carrier-bearing spectral projectors. The boundary incidence object is the physical isomorphism class of

$$\mathbf{B}_{\text{sec}}(X) = \left(F_\alpha \xrightarrow{B_{\alpha i}} P_i, F_\alpha \xrightarrow{B_{\alpha H}} P_H \right),$$

with light block $c_E U$ and heavy block $-s_E U$, modulo lawful rephasing, degenerate-heavy rotation, and boundary-preserving source isomorphism.

Theorem 9.1 (P6-08 – boundary-to-sector incidence). *For every ready object X , the complete charged-current, source, detection, conjugation, population, and symmetry boundary incidence descends to $\text{Sect}_{\text{act}}^b(X)$ and is covariant under lawful morphisms. No globally transported flavor label is introduced as an interior carrier identity.*

Proof. PCR-2 reconstructs flavor as a boundary relation and proves covariance of the target-facing interface. Restricting the complete one-apex incidence diagram to sector subobjects gives the displayed maps. Rephasing and source-basis changes act simultaneously on the incidence data and preserve their physical isomorphism class. Since the interior sector definition uses propagation, spectral, and source support rather than flavor names, no boundary label is promoted to primitive interior identity. $\square$

9.2 Continuation generators

The within-realization continuation category contains:

1. propagation and relative-phase continuation;
2. source/detector and boundary reconstruction;
3. threshold and oriented matching continuation;
4. RG/EFT continuation at the target-retained light scope;
5. D1/D4 common-scale continuation;
6. D4 population/background and stress-energy continuation;
7. D2 conjugation/LNV response continuation;
8. D5 symmetry, protector, breaking, and charge-lattice continuation where applicable;
9. no-repair deletion and failure continuation;
10. lawful source, gauge, basis, chart, and degenerate-subspace redescription.

Theorem 9.2 (P6-09 – coherent continuation category). *For each $b \in \mathcal{B}_6^{\text{ready}}$, the target-authorized within-realization continuations form a fail-closed category $\mathcal{U}_{\text{term}}^b$ acting on sectors, aggregates, relation objects, and the parent carrier core. The action is closed under composition, natural under restriction, and preserves the D3/D4/D5 target boundaries.*

Proof. The E-H7 package provides the joint continuation monoid and target-admitted witnesses. Internal restrictions of those continuations are defined when the relevant sector support is retained; otherwise the action receives a typed undefined or failure value. PCR-5C composition and path-equivalence guarantee coherence on the accepted child target, while the ARC and D4 boundaries forbid transport into D3-1.1 or D4-complete. Composition is therefore total on the fail-closed status object. $\square$

9.3 Threshold-labelled standing

The heavy threshold H_λ is retained in the matching channel while the target-retained light continuation is common. This separation is essential: it makes the heavy sector internally standing without making its three degenerate basis vectors distinct.

10 Complete terminal witness interface

10.1 Terminal semantic uses

Terminal identity concerns the ability of a subobject to carry the complete child role under future use. The terminal semantic-use trace records:

Definedness, TargetTyping, BoundaryReconstruction, Persistence, Propagation, Matching, RG/EFT, Population, Symmetry, Conjugation, Compatibility, Failure, Deletion, Locality, Representation, Charge, Constituent, Reuse, Redescription.

Definition 10.1 (Complete terminal signature). For a sector, aggregate, relation, or parent object A , define

$$\Sigma_{\text{term}}^b(A) = (w(u(A)))_{(w,u) \in \mathcal{W}_{\text{term}}^b \times \mathcal{U}_{\text{term}}^b}.$$

The signature takes values in a product of physical outputs and typed status objects, so partial, undefined, ineligible, failed, and update-bearing coordinates remain distinct.

Theorem 10.2 (P6-10 – terminal carrier-interface activation). *For each ready branch, the witness family $\mathcal{W}_{\text{term}}^b$ and continuation category $\mathcal{U}_{\text{term}}^b$ are complete for same-scope terminal carrier identity: every difference capable of changing terminal definedness, compatibility, failure, physical kind, or identity-preserving reuse changes at least one coordinate of Σ_{term}^b .*

Proof. The physical-nature target was formed before sector inspection. Its semantic use trace contains every inherited child-target use and the new terminal bearer uses. The relative completeness theorem [Theorem 5.5](#) proves that all such uses descend through the complete PCR-3/PCR-5E carrier and incidence trace together with the sector trace. Anchor changes leave the fixed child target; tensor changes alter one of the listed target-authorized consequences; and skin changes alter none. By the fixed-domain exhaustion and SCIC activation theorem, the induced witness family is complete once all tensor consequence, compatibility, failure, and continuation loci are represented. E-H1–E-H9 and the constructions of [Sections 5 and 9](#) provide exactly those loci. No additional same-scope classifier remains. $\square$

Status 10.3 (Activation status).

$$\text{ActivationCertified}_{\text{term}}^{b_{\text{acc}}}, \quad \text{ActivationCertified}_{\text{term}}^{b_{\text{glob}}}.$$

The certification is child-qualified and does not extend to the PCR-5E parent.

11 Terminal signatures of sectors, aggregates, relations, and parent

11.1 Signature and role-completeness calculus

For a candidate object A relative to a fixed accepted realization X , three questions must remain separate:

1. whether the candidate is well formed and all terminal witness coordinates are defined;
2. whether it is eligible for the independently formed complete-bearer role;
3. whether its complete continuation signature is the same as, or different from, the parent signature.

Accordingly write

$$\text{TotStat}_b(A) \in \{\text{Total}, \text{Partial}, \text{Failed}, \text{Undefined}, \text{TargetUpdate}\},$$

$$\text{RoleStat}_b(A) \in \{\text{RoleComplete}, \text{RolePartial}, \text{Failed}, \text{Undefined}, \text{TargetUpdate}\},$$

and, when the complete terminal signature is defined,

$$\text{IdRel}_b(A; X) \in \{\text{ParentEqual}, \text{ParentDifferent}\}.$$

Role-completeness is a capability and compatibility statement relative to the branch target. It does not mean equality with the particular parent values. Parent equality is an identity-stage comparison made only after eligibility and physical completeness have been formed.

11.2 Node and aggregate signatures

Let

$$\mathbf{S}_L = \mathbf{S}_1 \oplus \mathbf{S}_2 \oplus \mathbf{S}_3.$$

The light aggregate carries the oscillation, D1, and light D4 response. The heavy sector carries the D3 threshold, active-closure deficit, and D5 vectorlike matching incidence. Their complete child work is joined by the source-connected mass map, common charged-current boundary, matching, population, and symmetry relations.

Theorem 11.1 (P6-11 – sectorwise terminal continuation signatures). *For every $b \in \mathcal{B}_6^{\text{ready}}$ and $X \in \text{PhysMod}_{E,VJ}^b$:*

1. *each light sector $\mathbf{S}_1, \mathbf{S}_2, \mathbf{S}_3$ has a defined but role-partial terminal signature;*
2. *every proper light aggregate has a defined but role-partial or failed signature;*
3. *the full light aggregate $\mathbf{S}_L$ has a defined but role-partial signature;*
4. *the heavy sector $\mathbf{S}_H$ has a defined but role-partial signature;*
5. *every proper aggregate of $\{\mathbf{S}_1, \mathbf{S}_2, \mathbf{S}_3, \mathbf{S}_H\}$ is not role-complete.*

Proof. A single light sector lacks the no-repair multi-frequency oscillation packet, the complete D1 endpoint measure, the three-slot D4 population record, and the D3 heavy response. Deleting any light projector from a larger light aggregate changes the retained oscillation phase spectrum, the endpoint weight sum, and the D4 compiled light-population packet without retuning; hence every proper light aggregate fails at least one target use. The full light aggregate supplies the complete light-sector work but has no heavy threshold, active-closure completion, or vectorlike D3/D5 matching record. The heavy sector supplies those heavy records but has no oscillation, light endpoint, or cosmological light-population response. Any proper light-heavy aggregate omits at least one of these indispensable supports. Therefore no proper spectral aggregate is terminal-role-complete. $\square$

11.3 Relation signatures

The relation decoration contains:

- charged-current boundary incidence $\mathbf{B}_{CC}$;
- light-heavy mass and matching incidence $\mathbf{R}_{LH}$;
- active-closure and process incidence $\mathbf{R}_U$;
- D4 population/background relation $\mathbf{R}_P$;
- D5 accidental or exact-global action/protector relation $\mathbf{R}_G^b$;
- D1/D4 common-scale relation $\mathbf{R}_{\beta\Sigma}$;
- accidental LNV relation $\mathbf{R}_V(z)$ when source-active;
- the common-cone relation $\mathbf{R}_\Lambda$.

Lemma 11.2 (No relation-only terminal bearer). *Every proper relation object has a role-partial or formation-ineligible terminal status. No relation possesses independent persistence, complete boundary reconstruction, and all child-target continuations without its carrier-support nodes.*

Proof. Each relation is typed by its incident bearer support. Removing that support either makes the relation undefined or leaves it without persistence and boundary use. The matching relation has no independent propagation; the population relation has no carrier boundary; the symmetry action has no independent matter persistence; the LNV amplitude is a process leg rather than a complete carrier; and the common cone is a compatibility witness. Hence none can independently instantiate the complete-bearer role. $\square$

11.4 Target-derived terminal eligibility

For each ready branch, let

$$R_{\text{term}}^{\nu,b} = \{r_{\text{bear}}^b\}$$

be the one-role terminal architecture already fixed in [Theorem 4.2](#). The role r_{bear}^b is a capability bundle derived from $T_{\text{nat}}^{\nu,b}$: it requires same-child source typing, persistent neutral-fermion bearer support, the charged-current boundary interface, propagation and continuation capability, matching/RG–EFT capability, availability of the D1–D5 role blocks appropriate to branch b , one-apex formation, physical-kind admissibility, and absence of a source-level failure or update. It does not contain the numerical or categorical value of the parent continuation signature.

Theorem 11.3 (P6-S2A – target-derived terminal eligibility formation). *For every ready branch b , the eligibility relation*

$$E_{\text{term}}^b \subset \text{CandTerm}_b(X) \times R_{\text{term}}^{\nu,b}$$

is generated before terminal assignment by the branch target and source certificate alone. A positive edge

$$E_{\text{term}}^b(A, r_{\text{bear}}^b) = 1$$

requires exactly the following independently fixed conditions:

1. *A is a source-authorized same-child candidate with persistent bearer support;*
2. *the required boundary, propagation, continuation, matching, and physical-kind interfaces are well typed on A;*
3. *the branch-required D1–D5 role capabilities are available on the same physical apex;*
4. *no source equation, target applicability condition, or fixed failure predicate excludes the use; and*
5. *the eligibility edge is invariant under lawful source, basis, gauge, chart, and presentation redescription.*

The definition of E_{term}^b does not compare $\Sigma_{\text{term}}^b(A)$ with $\Sigma_{\text{term}}^b(X)$, and does not use parent SCIC identity as an eligibility premise.

Proof. The terminal role is formed in [Theorem 4.2](#) from the accepted child target before any candidate bearer is counted. Its formation data are the target roles, source sorts, boundary records, continuation capabilities, physical-kind requirements, and failure/update conditions imported through the E-H certificate and the physical-nature target. These are exactly the certificate-typed sources permitted by Role-Occupancy Closure: internal role typing, predeclared source/provenance typing, and stable boundary or role traces. Likelihood preference, source-family naming, a canonical representative, and complete-signature equality with the parent are not formation rules for eligibility. Lawful-redescription invariance follows from P6-S0, P6-08, P6-09, and terminal activation. Hence the relation is independently formed and target-derived. $\square$

Remark 11.4 (Negative control from PCR-5). A one-role target and a positive eligibility edge do not imply carrier identity. PCR-5 supplies a direct negative control: several SCIC-distinct carrier classes are eligible for the same complete-carrier role and the resulting terminal diagnosis is over-residue [\[Mal26c\]](#). P6-S2A therefore preserves, rather than forbids, the logical possibility of several eligible bearer identities. Any one-bearer conclusion in PCR-6 must come from the subsequent source-support and compatibility calculation.

11.5 Parent signature

Define the source-connected carrier core

$$C_{\nu}^{\text{VJ}}(X) = \text{ActRed}\left(S_1 \cup S_2 \cup S_3 \cup S_H \cup B_{\text{CC}} \cup R_{LH} \cup R_U \cup R_P \cup R_G^b \cup R_{\Lambda}\right),$$

with $R_{\beta\Sigma}$ and the active branch-specific relation decoration attached at their target-defined loci.

Theorem 11.5 (P6-12 – parent and cross-sector terminal signature). *The parent object $C_{\nu}^{\text{VJ}}(X)$, together with its target-defined standing decoration, has a complete terminal signature on one physical apex. Every terminal witness is localized to a carrier-support sector or a deletion-essential relation, and no witness is imported from a different realization or counted twice.*

Proof. PCR-5E supplies one source-faithful physical apex on which all inherited, marginal, pairwise, higher-arity, background, matching, continuation, and common-cone residuals vanish. Restricting that apex to the carrier core preserves the intrinsic vectorlike mass map, all four carrier-bearing spectral sectors, the common boundary, heavy/light matching, and the target-essential relation packet. D1/D4, D2, D3, D4, and D5 terminal uses are therefore all defined. The semantic saturation theorem assigns each joint witness to its actual incidence and prevents marginal splicing. Relation supports are disjoint as typed uses even when they share bearer nodes, so no witness is double counted. Thus the parent signature is complete. $\square$

11.6 Same-apex complete-bearer exhaustion

Let

$$K_{\min}^b(X)$$

denote the deletion-essential bearer support of the fixed realization: the three light sectors, the heavy sector, the charged-current boundary, active light-heavy incidence, D3/D5 matching and active-closure incidence, the D4 population/background relation, the branch-required D5 action or protector, and the common-cone relation, together with any target-active decoration whose deletion changes terminal standing. This support is generated from the source and deletion matrices, not from a preferred bearer label.

Theorem 11.6 (P6-S2B – same-apex complete-bearer exhaustion and parent-support uniqueness). *Fix $b \in \mathcal{B}_6^{\text{ready}}$ and $X \in \text{PhysMod}_{E,VJ}^b$. Let $A \hookrightarrow X$ be a source-authorized candidate with*

$$E_{\text{term}}^b(A, r_{\text{bear}}^b) = 1$$

and suppose A is terminal-role-complete and physically compatible on the fixed apex. Then:

1. *A contains every support and relation in $K_{\min}^b(X)$;*
2. *every proper omission of $K_{\min}^b(X)$ is role-partial, failed, undefined, or incompatible;*
3. *after removal of skin, provenance, and lawful presentation redundancy, A is the source-connected parent core $C_{\nu}^{\text{VJ}}(X)$;*
4. *consequently*

$$\Sigma_{\text{term}}^b(A) = \Sigma_{\text{term}}^b(C_{\nu}^{\text{VJ}}(X)) = \Sigma_{\text{term}}^b(X).$$

A full-support presentation with a different terminal signature cannot be a source-faithful subobject of the fixed realization: it is either a different PCR-5E realization class or a declared target/source update.

Proof. By P6-S0 and P6-04, every same-child candidate bearer has a source-generated normal form inside the activity-reduced subobject/relation category of X . P6-11 shows that every proper spectral node or aggregate omits terminal work. The relation-only theorem excludes every relation as an independently complete bearer. The no-repair deletion matrix further shows that removal of any light sector, the heavy sector, the light-heavy source incidence, the charged-current boundary, the D4 population/background relation, the branch-required D5 action or protector, or the common-cone relation changes a mandatory terminal witness or makes a continuation undefined. Hence a terminal-role-complete compatible candidate must contain all of $K_{\min}^b(X)$.

Now let A contain that full support. The full-support inheritance proposition [Theorem G.2](#) applies: a same-apex source-faithful subobject inherits the source parameters, target attachments, orientations, boundary data, continuation laws, and active incidence values of X ; subobject formation permits retention and deletion, not retuning. Any remaining difference is lawful presentation/skin or target-inert enrichment and collapses under the complete carrier trace. Changing a standing-bearing source or continuation value instead defines another joint realization X' or an explicit update. Therefore every independently eligible, compatible, terminal-complete candidate inside the fixed realization is lawfully equivalent to $C_{\nu}^{\text{VJ}}(X)$ and has the same complete continuation signature. $\square$

Corollary 11.7 (Complete bearer class inside a fixed realization). *The independently eligible candidates may be more numerous than one, but the subset that is simultaneously terminal-role-complete and compatible on the fixed apex has exactly one terminal SCIC class:*

$$\pi_0 \mathfrak{C}_{\text{live}}^b(X) = \{[C_{\nu}^{\text{VJ}}(X)]_{\text{term}}\}.$$

This singleton is a consequence of source-support exhaustion and one-apex compatibility, not an eligibility convention.

12 Promotion to terminal carrier identity

12.1 Promotion criterion

Definition 12.1 (Terminal promotion). An internal object A is promoted to a terminal bearer only if:

1. it is independently eligible for a role in the target-derived terminal role architecture;
2. it is terminal-role-complete on the fixed physical apex;
3. it participates in at least one jointly compatible complete assignment;
4. it remains identity-bearing under every licensed terminal continuation;
5. it is not merely a representative, projection, source label, or provenance tag.

Complete-signature equality with the parent is not an eligibility premise. It is a consequence only if the source-support exhaustion theorem shows that the promoted candidate is the fixed parent bearer up to lawful redescription.

Theorem 12.2 (P6-13 – terminal promotion necessity). *Target-derived terminal eligibility, terminal-role completeness, continuation stability, and assignment compatibility are necessary conditions for promotion of a sector, aggregate, relation, or parent object. Within a fixed ready-child realization, P6-S2B further implies that any candidate satisfying all of these conditions is lawfully equivalent to the source-connected parent bearer and therefore has the parent terminal continuation signature.*

Proof. A promoted object is proposed as a standing occupant of the independently formed terminal role. Role-Occupancy Closure therefore requires an eligibility edge and membership in a compatible complete assignment; stable reference requires witness soundness and completeness; reuse requires continuation stability; and the no-second-classifier theorem excludes promotion by label or preference. P6-S2A supplies eligibility without using parent identity. P6-S2B then proves, from source completeness, deletion-essential support, and one-apex compatibility, that every independently eligible and terminal-role-complete same-apex candidate contains the full parent support and is lawfully equivalent to the parent. Parent-signature equality is therefore derived after promotion conditions are met rather than inserted into their definition. $\square$

12.2 Spectral and representational routes

The following data are standing internal content or downstream presentations but are not sufficient carrier grounds:

- distinct mass or pole values;
- spectral projectors or eigenvalue labels;
- the rank-three heavy degeneracy multiplicity;
- the D1/D4 common absolute-scale coordinate τ ;
- the heavy threshold H_λ by itself;
- exact or accidental $B - L$ labels;
- charged-current boundary labels;
- Higgs, Yukawa, mass-matrix, diagonalization, or PMNS presentations.

Theorem 12.3 (P6-14 – primitive-carrier/spectral-sector nonpromotion). *No carrier-bearing spectral sector, metric image, symmetry label, boundary label, or matrix/eigenvalue presentation is promoted to a separate terminal carrier merely by being target-visible or numerically distinct. Promotion is possible only after independently formed eligibility, terminal-role completeness, continuation stability, and compatible Role-Occupancy Closure.*

Proof. By Theorem 11.1, every proper spectral sector or aggregate is terminal-role-incomplete. By Theorem 11.2, relation labels have no independent terminal bearer use. The D1/D4 τ -direction is an enriched standing coordinate with an exact fixed-base reduct; it changes response values but not the intrinsic source-connected carrier topology. Heavy threshold and symmetry data distinguish internal sectors and target decorations, but the associated proper sectors remain incomplete. Higgs/Yukawa/matrix/mixing objects are downstream representations whose identity-bearing content factors through already-standing carrier and response structure [Mal26q; Mal26s; Mal26aj; Mal26h; Mal26ak]. None of these distinctions therefore satisfies the independently formed promotion burdens. If a future same-scope candidate did satisfy them, P6-S2B rather than a naming convention would decide whether it is a second bearer or the parent presentation. $\square$

Corollary 12.4 (No three-masses/three-carriers inference). *The three nondegenerate light standing sectors do not imply three terminal carriers. They remain internal sectors of the terminal parent bearer proved below.*

Corollary 12.5 (No six-eigenvectors/six-carriers inference). *The six displayed eigenvectors of the block-diagonal presentation yield four standing spectral sectors and one terminal bearer, not six terminal carriers.*

13 Positive one-carrier realization

13.1 Source-connected parent core

At $s_E > 0$, the active Yukawa and Dirac-mass incidences connect the light and heavy source supports. The intrinsic source hypergraph is therefore connected after activity reduction. The exact block factorization does not split the physical source into independent terminal objects; it diagonalizes the response operator within one source-connected parent architecture.

Construction 13.1 (Parent terminal bearer). For $X \in \text{PhysMod}_{E, \text{VJ}}^b$, let $C_V^{\text{VJ}}(X)$ be the activity-reduced source-connected neutral-fermion carrier core with:

1. light and heavy spectral support;
2. canonical kinetic and persistence structure;
3. common charged-current boundary incidence;
4. active light-heavy source and mass-map incidence;
5. heavy threshold and oriented matching;
6. all deletion-essential target relations required by $T_{\text{nat}}^{\nu, b}$.

The D1/D4, D2, D4 population, and D5 symmetry values are retained in the standing decoration of the same carrier object and are not used to multiply its bearer nodes.

Theorem 13.2 (P6-15 – source-faithful positive one-carrier realization). *For every $b \in \mathcal{B}_6^{\text{ready}}$ and every $X \in \text{PhysMod}_{E, \text{VJ}}^b$, the parent object $C_V^{\text{VJ}}(X)$ is a source-faithful physical terminal bearer with*

$$\text{Res}_E^b(C_V^{\text{VJ}}(X)) \simeq X, \quad \chi_{\text{car}}(C_V^{\text{VJ}}(X)) \simeq \chi_{\text{car}}^{\text{parent}}(X),$$

and all terminal residuals vanish on one physical apex. Its internal carrier-bearing sector object is $\text{SpecCore}^b(X)$, and its complete terminal signature is $\Sigma_{\text{term}}^b(C_V^{\text{VJ}}(X))$.

Proof. The carrier core is obtained by restricting the accepted PCR-5E source/physical object to the source-connected matter bearer and all relations whose deletion changes terminal use. It retains the exact intrinsic vectorlike mass map, canonical kinetic terms, common boundary, active closure, threshold/matching, and target-required attachments. PCR-5E one-apex physicality proves that these data coexist on one semantic object. The restriction does not splice marginal witnesses and does not retune any source parameter. Theorem 11.5 gives complete terminal signature, and the source-normal-form theorem gives source fidelity. Therefore the constructed parent is a physical bearer, not merely a quotient label. $\square$

13.2 Deletion witnesses

The positive theorem is strengthened by no-repair deletion:

- deleting any light sector changes the exact oscillation/D1/D4 packet;
- deleting the heavy sector destroys the D3/D5 threshold and closure packet;
- deleting the light-heavy incidence disconnects the active source and invalidates the mass-map/matching trace;
- deleting the child symmetry relation fails the child D5 target;
- deleting the D4 population relation fails the public-summary child target;

- deleting the optional accidental LNV relation changes the enriched D2 class but leaves the minimal carrier core and target survival intact at the zero subfibre.

Proposition 13.3 (Parent minimal support at the bearer level). *The unique minimal carrier-support set is the full four-sector spectral core together with the boundary, light-heavy matching, active-closure, population, and child symmetry relations. Optional LNV response is not part of the common minimal bearer core.*

Proof. Necessity of the listed supports follows from the deletion failures above. Sufficiency follows from the one-apex realization theorem. The accidental LNV source is optional because its no-repair deletion reaches the accepted zero-response subfibre without changing the remaining source equations. $\square$

14 Exhaustion of alternative bearer architectures

14.1 Higher-arity bearer alternative

The source contains essential higher relations, including protected matching and common light-mass/population incidence. Their essentiality does not make them terminal bearers unless they independently carry persistence, boundary reconstruction, and the complete terminal signature.

Theorem 14.1 (P6-16 – higher-arity terminal bearer alternative). *No irreducible relation in either ready child is a standalone terminal bearer. Every deletion-essential higher relation is a tensor incidence of $C_v^{vj}(X)$ and enters its minimal identity packet. The higher-arity branch therefore reduces to the one-carrier parent branch rather than defining a distinct relational bearer.*

Proof. The relation signatures are ineligible or partial by [Theorem 11.2](#). Each relation depends on incident matter support, lacks independent boundary reconstruction or persistence, and becomes undefined when its carrier nodes are removed. Yet deleting the relation from the parent changes terminal use, so it is not skin. Anchor–tensor–skin descent therefore classifies it as tensor content of the parent carrier. No independent relation occupant remains. $\square$

14.2 Carrier-node separation exhaustion

A complete bearer could in principle be an individual sector, a proper aggregate, the parent, a relation, a mixed promoted subset, or no object at all. These are structural support modes, not an informal list of preferred models.

Theorem 14.2 (P6-17 – carrier-node separation exhaustion). *For every accepted ready-child realization, the total terminal disposition is*

$$\text{TermDisposition}_b(X) = \text{OneCarrier}.$$

No multi-carrier, higher-arity-carrier, mixed, inconsistent, or NotReady branch survives at the declared child scope.

Proof. Individual sectors and proper aggregates are excluded by [Theorem 11.1](#); relation bearers are excluded by [Theorem 14.1](#); the parent is complete by [Theorem 13.2](#); and activation is complete by [Theorem 10.2](#). For any finer source-authorized candidate not represented by the displayed macroblocks, P6-S2A forms eligibility independently and P6-S2B performs the source-support exhaustion. A candidate that could appear in a compatible complete assignment must contain the full deletion-essential parent support; proper omissions are incomplete, and full-support same-apex presentations reduce to the parent under lawful redescription. A mixed branch would require at least one promoted proper component, but none exists. Inconsistency is excluded by the accepted one-apex parent realization. The list is exhaustive because elementary same-scope support variation is anchor, tensor, or skin, and composite support is node, aggregate, relation, or parent. An alleged fifth bearer locus either changes target, duplicates one of these support modes, or carries no standing. $\square$

14.3 Comparison-qualified substitute classes

The ready E-H2 comparison category contains the source-qualified vectorlike-joint normal forms and their lawful presentations. Law-only, topological, composite, alternative hidden, and other PCR-3 parent architectures are not members of this completed child category and are not excluded by PCR-6.

Theorem 14.3 (P6-19 – comparison-qualified substitute finality). *Within each ready E-H2 child source category, every complete terminal presentation is lawfully equivalent to the source-connected vectorlike-joint parent carrier core. No second physical kind or source-distinct terminal bearer survives inside the child comparison class. Architectures in the unresolved parent complement retain their prior NotReady or ROOS status and are not excluded.*

Proof. E-S3–E-S5 freeze the raw and solved vectorlike source normal forms and their lawful groupoids before joint evaluation. Every accepted child object has that source fingerprint. Source/basis/gauge/chart differences within the groupoid preserve the complete carrier trace and therefore collapse by SCIC. More generally, P6-S2A permits all target-derived eligibility edges without imposing parent identity, while P6-S2B shows that any eligible presentation that is also terminal-role-complete and one-apex compatible must contain the full parent support and reduce to the fixed parent trace. A source form with different primitive carrier inventory lies outside E-H2 and belongs to the parent comparison problem. Thus kind finality is exact within the child and silent about the parent complement. $\square$

Status 14.4 (Scope of substitute exhaustion).

$$\text{KindFinality}_{\text{VJ}}^{b_{\text{acc}}}, \quad \text{KindFinality}_{\text{VJ}}^{b_{\text{glob}}}, \quad \text{ParentKinds} = \text{NotReady}_{\text{PCR5E}}.$$

15 Physical kind and locality of the terminal bearer

15.1 Kind-facing trace

The kind-facing trace records support, persistence, derivation, constituent structure, locality, representation, boundary relation, and higher-incidence status without using familiar model names as classifiers.

Definition 15.1 (Carrier kind classes). The candidate kind classes are existing-substrate/law, point-local matter, hidden/dilation, composite/emergent, topological/defect/extended, boundary/environmental co-carrier, hybrid, and irreducible relational carrier.

Theorem 15.2 (P6-18 – carrier-kind identification). *For every ready-child realization, the terminal bearer has the exact kind*

$$\text{Kind}(C_{\text{V}}^{\text{VJ}}(X)) = \text{LocalHybridNeutralFermion}_{\text{VJ}}.$$

Its intrinsic mass-map core is vectorlike Dirac; its light boundary is chiral; its complete source is a local hybrid of active and singlet neutral-fermion supports connected by source-active mass and charged-current incidence.

Proof. All intrinsic carrier fields have canonical local kinetic terms and appear in local gauge-invariant source operators. The bearer possesses point-local/effectively local persistence, activated Lorentz/spin-statistics structure, neutral fermionic representation, and a common charged-current boundary. It is not law-only because deleting the matter support destroys the target; not merely hidden because the light boundary is target-visible; not composite because no standing derivation from constituents replaces the primitive source support; not topological, defect, boundary, or environmental because its persistence and target path are local matter uses; and not relation-only by Theorem 14.1. It is hybrid because active and vectorlike singlet supports plus their irreducible incidence are jointly essential. $\square$

15.2 Locality, Lorentz covariance, and spin-statistics

Theorem 15.3 (P6-20 – locality and spacetime-presentation characterization). *The terminal carrier core is point-local at the intrinsic source level and admits the standard neutral spin- $\frac{1}{2}$ Lorentz presentation with fermionic statistics. D1 endpoint, D4 population/background, and detector/likelihood structures are target attachments and do not become the premetric carrier ground.*

Proof. The source operators and kinetic terms are local, the causal support and continuation witnesses are those of the accepted PCR-5E one-apex object, and the source fields are fermionic. The Lorentz/spin-statistics presentation is activated through the physical source attachment and preserves the premetric carrier trace. The D1/D4 and detector structures have exact forgetful maps to the same intrinsic source, so they are projection/response layers rather than carrier-identity generators. $\square$

15.3 Common carrier core

Define

$$\text{NatureCore}(X) = (\text{LocalHybridNeutralFermion}_{\text{VJ}}, \text{PointLocal}, \text{Fermionic}, \text{CCBoundary}, \text{Spec}_{3+1}, \text{LHMatch}, \text{ActiveClosure}, \text{ContinuationCore}).$$

This core is independent of the accidental/exact-global target tag. Target-specific symmetry and conjugation status enters the decoration defined later.

16 Gauge, charge, chirality, conjugation, and multiplicity

16.1 Quantum-number incidence graph

The intrinsic vectorlike source contains the active electroweak boundary fields, three active right-handed neutrinos, and equal-rank singlet multiplets N_L, N_R . The heavy pair is vectorlike, all neutral matter fields carry the declared conventional $B - L = -1$, and the complete source is anomaly compatible on the accepted D5 targets.

Theorem 16.1 (P6-21 – bearer/sector quantum-number packet). *For every ready-child realization, the terminal bearer carries the following standing quantum-number structure:*

1. a common neutral-fermion matter carrier with a chiral electroweak charged-current boundary;
2. a vectorlike heavy singlet sector with exact anomaly cancellation and one rank-three standing projector;
3. three light standing sectors with common bearer-wide electric neutrality and target-retained boundary incidence;
4. bearer-wide source covariance and common matching orientation;
5. branch-specific $B - L$ status: bounded accidental through $d \leq 4$ on b_{acc} , exact global with continuation-stable protector on b_{glob} ;
6. branch-specific conjugation/LNV response: optional accepted LNV relation on b_{acc} , defined zero protected response on b_{glob} .

No distinct charge or terminal carrier identity follows from a mass label alone.

Proof. The source coefficients are gauge invariant and the equal-rank heavy pair is vectorlike. D5 supplies the exact target-tagged action, anomaly, and protector data. Connected spectral charge uniformity shows that exact continuous/residual charge is constant on each connected spectral component, but does not itself close terminal identity. Terminal identity is supplied here by the complete parent signature and ROC. The accidental and exact-global differences remain target decoration because the underlying source-connected matter core and its terminal disposition are unchanged. $\square$

16.2 Multiplicity firewalls

The following quantities are distinct:

- algebraic heavy multiplicity: three degenerate heavy eigenvectors;
- heavy standing-sector cardinality: one;
- total carrier-bearing spectral-sector cardinality: four;
- terminal bearer cardinality: one;
- cross-realization child quotient cardinality: at least continuum.

Corollary 16.2 (Exact multiplicity tuple). *For every ready object,*

$$(\text{mult}_{\text{heavy}}^{\text{alg}}, |\pi_0 S_H|, |\pi_0 \text{SpecCore}^b(X)|, |\text{TermCar}_b(X)|, |Q_{E, \text{VJ}}^b|) = (3, 1, 4, 1, \geq c).$$

17 Standing coordinates and identity content

17.1 Imported PCR-5E incidence

PCR-5E proves the following relations at the completed child sectors:

$$\begin{aligned} D1/D4 &: \text{CommonGenerator}, & D2/D5_{\text{glob}} &: \text{ObstructionCoupled}, \\ D3/D5_{\text{acc,glob}} &: \text{IndependentCoordinate}, \end{aligned}$$

with no all-phenomenon common generator and no emergent joint-only carrier selector.

Theorem 17.1 (P6-R3 – standing-coordinate inheritance and fixed-base τ firewall). *PCR-6 preserves the PCR-5E coordinate-incidence relations. In particular, the common D1/D4 coordinate τ changes the enriched absolute-support and mass-sum signatures while preserving all seven PCR-5B base coordinates and the intrinsic vectorlike carrier core. It is a standing carrier-indexed attribute, not a new bearer node.*

Proof. E-S6 identifies the canonical D1/D4 common trace quotient, and E-S7A/E-S7 prove the exact fixed-base reduct and equality of all seven base packet channels along the τ -leaf. The intrinsic mass map, source topology, heavy threshold, boundary, activity, matching orientation, and failure frontier are fixed. Therefore τ changes the complete child SCIC class but not the within-realization bearer partition. $\square$

17.2 Classification of standing coordinates

Definition 17.2 (Coordinate roles). A target-visible coordinate is classified as one of:

BearerCore, InternalSector, EssentialRelation, StandingDecoration, EnrichedFibre,
Projection, Representative, NotReady.

Theorem 17.3 (P6-22 – operator, coupling, mass/deformation, and target-coordinate status). *The ready-child coordinates have the following exact roles:*

Coordinate	Role	Reason
Light/heavy source-connected support	BearerCore	Deletion destroys persistence and complete target work.
Three light projectors	InternalSector	Distinct relative propagation/boundary traces; terminal-incomplete individually.
Heavy rank-three projector	InternalSector	Threshold/matching standing; degenerate basis labels collapse.
Charged-current and light-heavy matching	EssentialRelation	Deletion changes terminal continuation; no independent bearer signature.
Active-closure coordinate s_E	Standing decoration; essential relation	Required by D3 child target but not a bearer partition.
Heavy threshold H_λ	Standing decoration; internal separator	Distinguishes joint classes and heavy sector; not a complete carrier.
Common scale τ	Enriched fibre; standing decoration	Changes D1/D4 child identity while preserving the base and bearer core.
D4 population/background	EssentialRelation	Required child target attachment; no independent matter persistence.
Accidental LNV amplitude z	Enriched fibre; optional relation	Changes D2 standing; deletion to zero preserves target survival and carrier core.
Exact-global protector	Standing decoration; essential relation	Required by exact-global target; not an independent carrier.
Higgs/Yukawa/mass matrix	Projection/representation	Represents already-standing response structure; no upstream authority.
PMNS/mixing chart	Boundary representation	Encodes invariant mismatch/boundary incidence; no interior flavor carrier.

Proof. Each classification follows from no-repair deletion, continuation, target restriction, and the imported common/independent/obstruction relations. Coordinates that alter the complete terminal use when deleted are retained in the identity decoration; coordinates whose variation changes only a downstream representation are projections; and none is promoted to a bearer without a complete terminal signature. $\square$

18 Role-Occupancy Closure within a fixed realization

18.1 The two different residue problems

PCR-5E applies Role-Occupancy Closure across the joint realization quotient. At that level, each child has one complete-carrier role and at least continuum many live joint carrier classes, so the diagnosis is over-residue. PCR-6 fixes one accepted realization X and forms a new within-realization residue from source-authorized candidate bearers of X . The two calculations use different candidate sets and must not be conflated.

Definition 18.1 (Within-realization terminal candidate class). Let

$$\text{CandTerm}_b(X)$$

be the activity-reduced, source-authorized class of candidate bearer subobjects, proper aggregates, irreducible relations, mixed supports, and the parent core inside the fixed realization, modulo presentation equivalence only. The class is generated before terminal eligibility or identity comparison.

18.2 Independent eligibility and assignment compatibility

For each branch the terminal role set is the target-derived one-role architecture

$$R_{\text{term}}^{\nu,b} = \{r_{\text{bear}}^b\}.$$

The eligibility relation E_{term}^b is the independently formed relation of P6-S2A. In particular, it is not defined by equality with $\Sigma_{\text{term}}^b(X)$.

The full assignment compatibility predicate is generated separately from the same target and physical certificate. For the one-role assignment $\sigma_A : r_{\text{bear}}^b \mapsto [A]$, define $J_{\text{term}}^b(\sigma_A; X) = 1$ exactly when all of the following hold:

1. A is source-faithful on the fixed physical apex;
2. A is terminal-role-complete;
3. the inherited boundary, matching, population, symmetry, and common-cone conditions are jointly compatible;
4. the licensed terminal continuations preserve the assignment; and
5. no applicable failure or undeclared target/source update occurs.

No parent-signature equality is included as a primitive compatibility clause.

Proposition 18.2 (Independent formation of the ROC data). *The tuple*

$$(R_{\text{term}}^{\nu,b}, E_{\text{term}}^b, J_{\text{term}}^b, \Gamma_{C_X, T}^{\text{lawful}})$$

is formed from the branch target, source category, physical apex, lawful-redescription groupoid, and continuation/failure structure before the assignment-orbit cardinality is evaluated.

Proof. The role set is fixed in Theorem 4.2. Eligibility is generated by P6-S2A from target typing, source provenance, stable boundary/role traces, and capability. Compatibility is the one-apex conjunction of source, boundary, continuation, matching, population, symmetry, failure, and terminal-completeness conditions already present in the target certificate. Lawful automorphisms are the automorphisms preserving this invariant tuple. None of these definitions consults the desired number of bearer classes. $\square$

Theorem 18.3 (P6-23 – within-realization terminal Role-Occupancy Closure). *For every $b \in \mathcal{B}_6^{\text{ready}}$ and every $X \in \text{PhysMod}_{E, \text{VJ}}^b$, the live within-realization terminal residue is*

$$\text{Residue}_{\text{term}}^{\text{live}}(X) = \{[C_{\nu}^{\text{VJ}}(X)]_{\text{term}}\}.$$

The compatible assignment set has one lawful orbit:

$$\left| \Sigma_{C_X}(T_{\text{term}}^{\nu,b}) / \Gamma_{C_X,T}^{\text{lawful}} \right| = 1.$$

Consequently

$$\text{RealizationLicensed}(C_X, T_{\text{term}}^{\nu,b}), \quad |\text{TermCar}_b(X)| = 1.$$

Proof. The eligibility relation is independent of parent identity by P6-S2A, so a second SCIC-distinct bearer is not excluded by definition. The uniqueness calculation is instead supplied by P6-S2B. Any candidate appearing in a compatible complete assignment must be independently eligible, terminal-role-complete, source-faithful, and one-apex compatible. P6-S2B then forces it to contain the complete deletion-essential support $K_{\min}^b(X)$ and to be lawfully equivalent to $C_V^{\text{VJ}}(X)$. Standing Carrier Identity Closure therefore puts every live candidate in the same terminal class.

The parent class itself is live: P6-12 and P6-15 prove its complete terminal signature, source fidelity, and one-apex physical realization, while P6-S2A supplies its eligibility. Hence the live residue is exactly the singleton parent class. The lawful automorphism group contains source/basis/gauge/redescription symmetries, including the heavy-degenerate $U(3)_H$ action, all preserving the complete role-incidence tuple. They do not produce a second assignment orbit. The Role-Occupancy Closure biconditional therefore licenses one bearer. $\square$

18.3 Separation of the two Role-Occupancy problems

Corollary 18.4 (One bearer does not collapse the child quotient). *For each ready branch,*

$$\forall X \in \text{PhysMod}_{E,\text{VJ}}^b : |\text{TermCar}_b(X)| = 1, \quad \text{while} \quad |Q_{E,\text{VJ}}^b| \geq c.$$

Thus one terminal bearer per realization is compatible with continuum many SCIC-distinct realizations.

Corollary 18.5 (Cross-realization over-residue remains final). *The PCR-5E status OverResidue is unchanged. PCR-6 supplies no selector among the continuum joint classes.*

Status 18.6 (Two-level ROC result).

$$\text{ROC}_{\text{cross}}^E(b) = \text{OverResidue}, \quad \text{ROC}_{\text{within}}^6(b, X) = \text{LicensedOneBearer}.$$

19 Minimal physical nature and target-relative identity

19.1 Two-layer packet

The exact target-relative identity of a child object contains more standing information than its physical kind. We therefore separate a common bearer core from branch- and point-dependent standing decoration.

Definition 19.1 (Common carrier-core packet).

$$\text{NatureCore}(X) = (\text{LocalHybridNeutralFermion}_{\text{VJ}}, \text{PointLocal}, \text{Fermionic}, \text{Neutral}, \text{VectorlikeMassCore}, \\ \text{CCBoundary}, \text{Spec}_{3+1}, \text{LightHeavyIncidence}, \text{ActiveClosure}, \text{MatchingContinuation}).$$

Definition 19.2 (Standing decoration). The branch-indexed decoration $\text{Dec}^b(X)$ contains all target-relative standing coordinates that are deletion-essential for the exact child identity but do not create new terminal bearer nodes. These include the heavy threshold/matching value, the D1/D4 common-scale trace, D4 population/background status, the child D5 action/protector, and any source-active accidental LNV response.

Definition 19.3 (Layered minimal identity).

$$\text{Identity}_{\min}^b(X) = \text{NatureCore}(X) \oplus \text{Dec}^b(X).$$

The first summand answers physical kind and carrier organization; the second preserves the exact target-relative SCIC identity.

19.2 Minimality of the identity packet

Theorem 19.4 (P6-24 – minimal identity and moduli naturality). *For every ready object X , $\text{Identity}_{\min}^b(X)$ is sufficient for every terminal use and minimal under no-repair deletion. Every retained field changes at least one terminal witness, continuation, compatibility, or failure value; every omitted source label, basis choice, degenerate-heavy vector label, matrix presentation, detector nuisance coordinate, and nonactive source cell changes none. The carrier-core packet is natural across each ready moduli groupoid.*

Proof. Sufficiency follows from the complete parent signature and one-apex realization. Minimality of the carrier support and essential relations follows from the deletion witnesses in Section 13. The exact child identity additionally requires the target-standing decoration: changing λ , τ , branch symmetry/protector status, or active z changes a complete target witness. Conversely, basis labels in the heavy degenerate subspace, source presentation, matrix coordinates, and inactive cells factor through lawful redescription or null reduction and can be removed without changing any terminal use. Naturality follows from Theorem 8.1: lawful moduli morphisms transport the common core and the corresponding decoration functorially. $\square$

Corollary 19.5 (P6-R2b – moduli-natural terminal nature packet). *For each ready branch, the assignments*

$$X \mapsto [\mathbf{C}_{\nu}^{\text{VJ}}(X)]_{\text{term}}, \quad X \mapsto \text{Identity}_{\min}^b(X)$$

extend over $\text{PhysMod}_{E,\text{VJ}}^b$ to a lawful terminal-bearer section and a moduli-natural minimal-identity packet. Every lawful morphism preserves the unique terminal assignment orbit, transports the four-sector core, and carries each target-standing decoration to its corresponding value.

Proof. P6-R2 transports the source-formed sector core and relation decoration. The source-support uniqueness theorem Theorem 11.6 and the within-realization ROC theorem Theorem 18.3 make the parent terminal class unique in every fibre, so lawful moduli morphisms cannot choose a different bearer class. P6-24 proves that the minimal packet is exactly the deletion-essential core plus target-standing decoration. Since those fields are natural under the imported PCR-5E restriction and continuation functors, the bearer class and minimal packet transport functorially. $\square$

19.3 Nature variation status

Proposition 19.6 (Exact-global nature variation). *On b_{glob} , the common carrier core and the qualitative standing-decoration type are constant across the ready moduli. Numerical values of m_0 , λ , τ vary inside the same packet schema.*

Proposition 19.7 (Accidental nature variation). *On b_{acc} , the common carrier core is constant. The full internal decoration is constructibly stratified by the LNV activity locus $z = 0$ versus $z \neq 0$, while the terminal bearer and its minimal core remain unchanged.*

Status 19.8 (Moduli nature status).

$$\text{NatureVar}_{b_{\text{glob}}} = \text{ConstantCore}, \quad \text{NatureVar}_{b_{\text{acc}}} = \text{ConstructiblyStratifiedDecoration}.$$

20 Cross-sector nature comparison

20.1 Comparison of the common source and carrier core

The two completed sectors share the same intrinsic vectorlike source normal form, parameter intervals, active mixing s_E , carrier-bearing spectral core, boundary, D3 closure packet, D4 public-summary attachment, and physical threshold/matching architecture. They differ only in the D2/D5 relation and its target-authorized symmetry/conjugation status.

Theorem 20.1 (P6-R4 – cross-child nature comparison). *For corresponding source points with the same (m_0, λ, τ) , the accidental and exact-global terminal bearers have lawfully equivalent common carrier cores:*

$$\text{NatureCore}_{b_{\text{acc}}}^b(X_{\text{acc}}) \cong \text{NatureCore}_{b_{\text{glob}}}^b(X_{\text{glob}}).$$

Their complete minimal identity packets are not identical:

$$\text{Identity}_{\min}^{b_{\text{acc}}}(X_{\text{acc}}) \neq \text{Identity}_{\min}^{b_{\text{glob}}}(X_{\text{glob}})$$

in general, because the exact-global child contains a continuation-stable protector and defined D2 zero subfibre, whereas the accidental child contains only bounded accidental $B - L$ and may carry an active LNV response. The exact cross-child status is

$$\text{CrossChildNature} = \text{CommonCarrierCore_TargetDependentDecoration}.$$

Proof. The E-S3/E-S4/E-S5 source certificates use the same vectorlike mass-map source and physical H/U/L realization for both children. The common carrier-support, boundary, spectral, closure, and matching traces therefore agree. The D5 target tags and D2 relation do not alter source connectivity or terminal bearer cardinality, so the carrier core is equivalent. However, deleting the exact-global protector fails b_{glob} , while the same structure is not required by b_{acc} ; conversely, an active accidental LNV response has no exact-global counterpart. These are target-standing differences and must remain in the complete identity decoration. $\square$

20.2 Branch stability across the completed sectors

Theorem 20.2 (P6-R5 – branch-stable one-carrier theorem). *Across both completed vectorlike-joint child sectors and throughout their accepted moduli,*

$$\begin{aligned} \text{TermDisposition}_b(X) &= \text{OneCarrier}, & |\text{TermCar}_b(X)| &= 1, \\ |\pi_0 \text{SpecCore}^b(X)| &= 4, & \text{Kind}(\text{C}_v^{\text{VJ}}(X)) &= \text{LocalHybridNeutralFermion}_{\text{VJ}}. \end{aligned}$$

The carrier core is common across children, while the exact minimal identity is target-decorated. This status is denoted

$$\text{BranchStableOneCarrier}_{\text{VJ}} \oplus \text{CommonCore_TargetDependentDecoration}.$$

Proof. Branchwise one-carrier closure is Theorems 14.2 and 18.3; exact four-sector structure is Theorem 7.4; moduli stability is Theorems 8.1 and 19.4; and common-core comparison is Theorem 20.1. Combining these results yields the stated branch-stable theorem without identifying the complete child packets or the PCR-5E joint realization classes. $\square$

Remark 20.3 (Why literal CommonNature is not claimed). The strongest skeleton branch CommonNature would require equality of the complete minimal child packets. The child target tags make that false: exact-global protection and accidental LNV permissibility are standing distinctions. The common carrier core is the strongest faithful cross-child conclusion.

21 Premetric-to-metric, Higgs, Yukawa, and mixing projection

21.1 Dependency direction

The dependency order is

$$\begin{aligned} \text{AASC governance} &\longrightarrow \text{C}_v^{\text{VJ}}(X) \longrightarrow \text{SpecCore}^b(X) \\ &\longrightarrow \text{mass/deformation response} \longrightarrow \text{metric and coupling presentations}. \end{aligned}$$

A mass matrix, diagonalization, Yukawa entry, Higgs vev, PMNS chart, or pole label can represent already-standing structure but cannot generate the terminal partition retroactively.

21.2 Projection functors

Define partial projection functors

$$\Pi_{\text{met}} : \text{Identity}_{\min}^b(X) \longrightarrow \text{MetricSpec}(X),$$

$$\Pi_H : \text{Identity}_{\min}^b(X) \longrightarrow \text{HiggsCouplingPres}(X), \quad \Pi_{\text{mix}} : \text{Identity}_{\min}^b(X) \longrightarrow \text{BoundaryMismatchPres}(X).$$

They are defined only where the relevant physical attachment and overlap witnesses exist.

Theorem 21.1 (P6-25 – metric/Higgs/Yukawa/spectral no-backflow). *The projection functors preserve the one-bearer identity and four-sector carrier-bearing spectral core. Distinct projected masses, poles, thresholds, Yukawa values, or mixing coordinates create no new premetric terminal bearer unless they already separate complete terminal signatures before projection. In particular:*

1. *the three light metric masses represent the three light internal sectors, not three terminal carriers;*
2. *the heavy triplet represents one standing heavy sector because its internal labels are related by the lawful degenerate-subspace action;*
3. *the τ -dependent D1/D4 metric responses distinguish joint SCIC classes but preserve the fixed carrier core;*
4. *Higgs/Yukawa presentations factor through the already-standing bridge/coupling/mass-response carriers;*
5. *PMNS-like data represent boundary mismatch and do not define globally transported flavor carriers.*

Proof. The premetric carrier and internal sectors have already been fixed by complete source and terminal signatures. MGN5 proves that matrix and eigenstructure stand only through an admitted mass-response carrier; MGN6 proves that mixing stands through invariant mismatch or boundary reconstruction; and the Higgs arc fixes Higgs identity and coupling/scale descriptions downstream of the bridge/fixation carrier [Mal26q; Mal26s; Mal26aj; Mal26h; Mal26w; Mal26n; Mal26al; Mal26ak]. The projections therefore factor through $\text{Identity}_{\min}^b(X)$. A projected distinction that changed terminal identity would have to correspond to an already-standing premetric tensor coordinate or else declare a target update. The fixed-base τ theorem supplies the explicit negative control. $\square$

Corollary 21.2 (Premetric carrier with multiple metric poles). *One premetric terminal bearer may project to several metric pole sectors without contradiction. PCR-6 realizes this structure as three light pole sectors plus one degenerate heavy pole sector.*

22 Main result

22.1 Summary of branchwise conclusions

Object	Status	Exact meaning
Completed sectors	$\mathcal{B}_6^{\text{ready}}$	D3-1.3/public-summary bounded-accidental and exact-global vectorlike-joint sectors.
Parent PCR-5E family	NotReady/ROOS	No completion, selection, or exclusion supplied by PCR-6.
Carrier-bearing spectral core	ThreeLightOneHeavy	Three nondegenerate light rank-one sectors plus one rank-three heavy sector.
Full internal sector object	Natural/stratified	Constant spectral core; accidental optional LNV relation stratifies at its activity frontier.
Terminal bearer count	One	One complete source-connected parent bearer in every ready realization.
Carrier kind	Local hybrid neutral fermion	Vectorlike Dirac mass-map core with chiral light boundary and essential matching relations.
Within-realization ROC	Licensed	One role, one live bearer class, one lawful assignment orbit.
Cross-realization ROC	OverResidue	PCR-5E continuum quotient remains non-singleton.
Cross-child nature	Common core, target decoration	Same carrier core; distinct exact-global/accidental symmetry-conjugation packet.
Metric/Higgs/Yukawa relation	No backflow	Downstream presentations preserve, not generate, terminal identity.

22.2 Principal theorem

Theorem 22.1 (PCR-6 branch-stable physical-nature and minimal-identity theorem). *Let $b \in \mathcal{B}_6^{\text{ready}}$ and $X \in \text{PhysMod}_{E, \text{VJ}}^b$. Then:*

1. X has exactly one terminal bearer $\mathcal{C}_V^{\text{VJ}}(X)$;
2. terminal eligibility is formed independently of parent SCIC identity, and every independently eligible candidate that is also terminal-role-complete and one-apex compatible reduces by source-support exhaustion to the parent terminal class;
3. the carrier-bearing spectral core has exact standing cardinality four,

$$\text{SpecCore}^b(X) = \{S_1, S_2, S_3, S_H\};$$

4. the three light sectors are nondegenerate rank-one sectors and the heavy sector is one rank-three degenerate sector modulo its lawful $U(3)$ action;
5. no proper sector, aggregate, or relation is terminal-complete;
6. the terminal bearer is a local hybrid neutral-fermion carrier with a vectorlike Dirac mass-map core;
7. the within-realization terminal Role-Occupancy assignment quotient is singleton;
8. the common carrier core is natural across the ready moduli and common to both child targets;
9. the complete minimal child identity packet is target-decorated, with accidental and exact-global symmetry/-conjugation structures remaining distinct;
10. the PCR-5E child quotient remains of cardinality at least continuum and the parent family remains unresolved;
11. metric, Higgs, Yukawa, pole, and mixing presentations have no backward carrier-authorizing force.

Therefore

$$\text{BranchStableOneCarrier}_{\text{VJ}} \oplus \text{ThreeLightOneHeavySectorCore} \\ \oplus \text{CommonCarrierCore_TargetDependentDecoration}$$

Proof. Item 1 follows from Theorems 11.3, 11.6 and 18.3; item 2 is Theorems 11.3 and 11.6; items 3 and 4 are Theorems 7.3 and 7.4; item 5 is Theorems 11.1, 11.6 and 14.1; item 6 is Theorems 15.2 and 15.3; item 7 is Theorem 18.3; item 8 is Theorems 19.4 and 19.5; item 9 is Theorem 20.1; item 10 follows from the two-level ROC firewall and Theorem 2.2; and item 11 is Theorem 21.1. The boxed status is their conjunction. $\square$

22.3 Strength and limitation of the result

The theorem attains the strongest bearer-cardinality conclusion available at the completed target scope on both branches and sharpens the internal-sector result from $r \geq 2$ to an exact $3 + 1$ carrier-bearing spectral stratification. The complete child identity packets are not identical, because accidental and exact-global target decorations remain standing-distinct. The strongest common conclusion is therefore equality of the carrier core together with target-dependent decoration.

Status 22.2 (Final PCR-6 status).

$$\text{Status}_{\text{PCR6}}(\mathcal{B}_6^{\text{ready}}) = \text{Complete}_{\text{child-qualified}}, \quad \text{Status}_{\text{PCR6}}(\mathcal{P}_E \setminus \mathcal{B}_6^{\text{ready}}) = \text{NotReady}_{\text{PCR5E}}.$$

23 Consequences for subsequent analysis

23.1 Target-qualified consequences

Theorem 23.1 (P6-26 – target-qualified downstream export). *PCR-6 exports, for each ready child:*

1. the exact terminal disposition and within-realization assignment orbit;

2. *the common carrier-core packet and branch-specific minimal identity decoration;*
3. *the four-sector carrier-bearing spectral core and full internal relation decoration;*
4. *the physical-kind, locality, quantum-number, and coordinate-role packets;*
5. *the projection firewall and exact parent nonpromotion boundary;*
6. *the moduli-naturality and cross-child comparison results;*
7. *the exact conditions under which any conclusion can be revised.*

No exported statement is universalized beyond the two ready child scopes.

Proof. Each item is an output of the completed theorem chain and retains its target/source tags. The parent firewall forbids broader quantification, and the no-backflow theorem forbids downstream observables or generation labels from strengthening the carrier conclusion without a new target. $\square$

23.2 Observable consequences

An observable analysis may now ask which consequences are common to every lawful realization in either completed sector. It must quantify over the continuum PCR-5E quotient and preserve the one-bearer/four-sector architecture rather than select a preferred (m_0, λ, τ, z) representative.

23.3 Relation to the fermion carrier-generation-sector programme

For the later fermion carrier-generation-sector analysis, PCR-6 supplies the following neutrino result:

$$\text{OneCarrier}_{\nu, \text{VJ}}, \quad \text{LightSectorCardinality} = 3, \quad \text{TargetSpecificHeavySector} = 1.$$

This does not yet prove that the three light sectors are the universal generation-sector realization for all fermions. That programme must independently construct the generation-to-sector correspondence and test charged/colored targets. The heavy VJ sector must be kept separate from the three light generation-like sectors.

Warning 23.2 (No universal neutrino export). The output is not $\forall X \in \text{NeutrinoUniverse} : |\text{TermCar}(X)| = 1$. It is the exact vectorlike-joint child theorem proved above.

24 Counterexample analysis and robustness

A relevant counterexample must preserve the completed child target, the E-H2 source comparison class, one-apex physicality, the activated terminal witness family, lawful equivalence, and the continuation category. A proposal that changes the source ontology or moves to an unresolved PCR-5E sector is an extension of scope rather than a counterexample to the theorem proved here.

Potential counterexample	Required burden and resolution
Target dependence on the desired result	Exhibit a carrier count, spectral rank, or preferred mechanism in target formation. P6-02 excludes this by target neutrality.
Incomplete physical-nature use class	Exhibit a same-sector bearer use that does not factor through the PCR-3/PCR-5E/sector trace. P6-S0 proves the required factorization.
Incomplete sector source theory	Exhibit a same-sector semantic subobject not reconstructed by the source theory. P6-S0 together with P6-03 and P6-04 closes this route at the finite E-H2 scope.
Incorrect activity reduction	Exhibit a nonzero incidence removed as null or a source-null spectator retained as active. P6-05 fixes activity through the source equations and attachment data.

Potential counterexample	Required burden and resolution
Incorrect light-sector count	Show that two light projectors have equal complete internal traces or that an additional light spectral root is target-visible. The nondegenerate light spectrum and trace exhaustion exclude both possibilities.
Incorrect heavy-sector quotient	Exhibit an activated propagation, boundary, D3, CLFV, matching, D5, failure, kind, or reuse witness that distinguishes heavy basis vectors while preserving exact degeneracy and lawful $U(3)_H$. P6-S1 proves descent for every such witness class.
Incomplete terminal activation	Exhibit a terminal use absent from the activated signature. P6-S0 and P6-10 derive the witness family from the complete semantic use class.
Nonphysical parent carrier	Show that the parent bearer is only a quotient label or cannot be realized on the accepted physical apex. P6-15 identifies it with the source-connected PCR-5E realization.
Circular eligibility	Show that terminal eligibility is defined through equality with the parent SCIC signature. P6-S2A forms eligibility independently of parent identity.
Second complete bearer	Exhibit an independently eligible, terminally complete, one-apex compatible candidate with full target support that is not lawfully equivalent to the parent. P6-S2B excludes this by deletion-essential support and same-apex source exhaustion.
Missed multi-carrier decomposition	Exhibit a proper component or finer source-authorized candidate that is independently eligible, complete, and compatible. P6-11 closes the displayed aggregates and P6-S2B closes the remaining source-authorized subobjects.
Missed relational bearer	Exhibit a relation with independent persistence, boundary reconstruction, and complete terminal signature. P6-16 excludes this class.
Incomplete physical-kind classification	Exhibit a complete kind within E-H2 outside the local hybrid vectorlike-joint normal form. P6-19 closes the child comparison class while leaving the unresolved parent source universe open.
Cross-sector overidentification	Show that the accidental and exact-global identity packets are treated as identical. P6-R4 preserves their target-dependent decoration.
Downstream identity generation	Show that a metric or Yukawa representation creates carrier identity absent from the terminal signature. P6-25 proves that every such distinction descends from standing premetric structure or changes target.
Confusion of Role-Occupancy levels	Infer a singleton PCR-5E quotient from one bearer, or infer bearer multiplicity from the continuum child quotient. The two-level Role-Occupancy theorem excludes both substitutions.
Parent-scope leakage	Derive a conclusion for D3-1.1, D4-complete, or another unresolved PCR-5E sector from the completed child theorem. P6-R1 preserves the parent boundary.

24.1 Logical structure of the one-carrier implication

The principal implication has the following form:

$$\begin{aligned}
 &\text{Activated} \wedge \text{SourceUseComplete} \wedge \text{ProperComponentsClosed} \\
 &\quad \wedge \text{EligibilityIndependent} \wedge \text{ParentSupportUnique} \\
 &\quad \wedge \text{ParentOneApexComplete} \wedge \text{ROC}_1 \implies |\text{TermCar}(X)| = 1.
 \end{aligned}$$

The premises are supplied by distinct theorems. In particular, P6-S2A forms eligibility without reference to the parent identity signature, while P6-S2B proves uniqueness from source completeness, deletion-essential support,

subobject inheritance, and one-apex compatibility. The one-carrier conclusion therefore does not follow from spectral multiplicity, source connectedness, or the cardinality of the target role set alone.

Status 24.1 (Robustness status). No faithful same-child counterexample is active after the complete theorem chain. Scope-changing parent or alternative-source proposals remain admissible future work and do not alter this status.

25 Conditions for extension

The theorem is relative to its stated target and comparison class. Its conclusions can change only through a new result that alters one of the premises below, not through a different interpretation of the same labels.

25.1 Conditions that would alter the completed-sector conclusions

A completed-sector conclusion would require revision if one of the following were established:

1. a faithful E-H2 source presentation with a semantic sector not covered by the sector grammar;
2. a target-authorized witness separating heavy degenerate basis vectors;
3. a proper sector, aggregate, or relation with a complete terminal signature and compatible role assignment;
4. a failure of the one-apex parent realization or source fidelity;
5. a lawful cross-child equivalence identifying the standing decorations previously distinguished;
6. a projection theorem showing that a downstream distinction changes terminal identity without an already-standing premetric tensor or target update.

25.2 Extension to additional PCR-5E sectors

Closure of additional PCR-5E sectors would enlarge $\mathcal{B}_6^{\text{ready}}$. Extending PCR-6 would then require a new branch-indexed completeness theorem, internal-sector classification, terminal activation, and cross-sector comparison. Existing child results would remain valid unless a new target explicitly changes their comparison class.

25.3 D4 and D3 boundaries

A complete D4 cosmological analysis requires JLR-D4-1. D3-1.1 requires its own Global Fibre Completion. Neither can be supplied by the present physical-nature theorem.

25.4 Extension to other fermion classes

Generalization to charged leptons and quarks requires the FCGS target system. Similarity of a three-light sector count is not sufficient; the source, boundary, confinement, charge, transition, and terminal-role architectures must be independently formed.

26 Conclusion

PCR-6 determines the physical nature and minimal terminal identity of the neutrino carrier in the two vectorlike-joint sectors completed by PCR-5E. The analysis separates three objects that ordinary notation can easily conflate: the continuum of joint realization classes across the PCR-5E quotient, the standing internal sectors of one accepted realization, and the independently complete terminal carriers within that realization.

The internal spectral structure is exact. Three nondegenerate light projectors remain distinct through their relative propagation and charged-current boundary traces. The three algebraic heavy eigenvectors belong to one standing sector because the complete activated witness family descends through the lawful $U(3)_H$ action on the degenerate

heavy subspace. The carrier-bearing spectral core is therefore

$$\text{SpecCore}(X) = \{S_1, S_2, S_3, S_H\},$$

with three light sectors and one rank-three heavy sector. Population, matching, symmetry, common-scale, and LNV structures remain target-relevant internal relations or decorations rather than independent carrier nodes.

Terminal carrier identity is obtained by a separate argument. Every proper sector, proper aggregate, and irreducible relation lacks some part of the complete terminal work. Eligibility for the bearer role is derived from the target independently of parent identity. Same-apex source-support exhaustion then shows that any eligible candidate capable of participating in a complete compatible assignment must contain every deletion-essential parent support. Proper omissions fail; full-support candidates inherit the fixed source and continuation data of the accepted realization and reduce to the source-connected parent carrier under lawful redescription. The live terminal residue consequently contains one carrier class, and Role-Occupancy Closure gives one compatible assignment orbit.

For every completed branch $b \in \mathcal{B}_6^{\text{ready}}$ and every accepted realization $X \in \text{PhysMod}_{E,VJ}^b$,

$$|\text{TermCar}_b(X)| = 1.$$

The terminal object is a local hybrid neutral-fermion carrier with a vectorlike Dirac mass-map core, common charged-current boundary incidence, and target-essential matching and continuation structure. Its carrier core is natural across both completed moduli. The bounded-accidental and exact-global sectors share that core but retain different complete identity packets because accidental LNV permissibility and exact-global protection are standing target data. The strongest cross-sector statement is therefore common carrier core with target-dependent decoration.

The theorem does not select one PCR-5E realization. Continuum many SCIC-distinct joint classes remain in each completed sector, and the PCR-5E cross-realization Role-Occupancy diagnosis remains over-residue. Nor does PCR-6 complete any unresolved parent sector or extend the result to all neutrino, charged-lepton, or quark architectures.

The final result is

$$\begin{aligned} &\forall b \in \mathcal{B}_6^{\text{ready}}, \forall X \in \text{PhysMod}_{E,VJ}^b : |\text{TermCar}_b(X)| = 1, \\ &|\pi_0 \text{SpecCore}^b(X)| = 4 = 3 \text{ light sectors} + 1 \text{ heavy degenerate sector}, \\ &\text{Kind}(\mathcal{C}_V^{\text{VJ}}(X)) = \text{LocalHybridNeutralFermion}_{VJ}, \\ &\text{CrossChildNature} = \text{CommonCarrierCore_TargetDependentDecoration}, \\ &|Q_{E,VJ}^b| \geq \mathfrak{c}, \quad \text{Status(PCR-5E parent)} = \text{NotReady/ROOS}. \end{aligned}$$

The neutrino result provides a precise calibration for the later fermion carrier-generation-sector problem: three light standing sectors can be internal to one terminal carrier, while an additional heavy sector remains part of the same vectorlike-joint bearer. Whether comparable carrier-sector structure is forced for charged leptons or quarks remains a separate target-relative theorem problem.

A Notation and status calculus

Symbol/status	Meaning
$\mathcal{B}_6^{\text{ready}}$	The two PCR-5E target-and-comparison-qualified vectorlike-joint completed child sectors.
$b_{\text{acc}}, b_{\text{glob}}$	Bounded-accidental and exact-global child tags.
$\text{PhysMod}_{E,VJ}^b$	Accepted one-apex physical groupoid of child b .
$Q_{E,VJ}^b$	PCR-5E child joint SCIC quotient across realizations.
X	One accepted vectorlike-joint physical realization.
$\text{Sect}_{\text{raw}}^b(X)$	Source-generated candidate internal sector presentations.
$\text{Sect}_{\text{act}}^b(X)$	Activity-reduced internal standing-sector groupoid.
$\text{SpecCore}^b(X)$	Carrier-bearing spectral subquotient.

Symbol/status	Meaning
S_1, S_2, S_3	Three nondegenerate light rank-one standing sectors.
S_H	One rank-three heavy degenerate standing sector.
$\text{RelDecor}^b(X)$	Boundary, matching, population, symmetry, conjugation, and higher-relation decoration.
$\Sigma_{\text{term}}^b(A)$	Complete terminal continuation signature of A .
$C_{\nu}^{\text{VJ}}(X)$	Source-connected parent terminal bearer.
$\text{TermCar}_b(X)$	Terminal bearer set after promotion and within-realization ROC.
$\text{NatureCore}(X)$	Common physical-kind and carrier-organization packet.
$\text{Identity}_{\min}^b(X)$	Layered minimal target-relative identity packet.
NotReady	Named readiness failure; not an exclusion and not a third standing value.
ROOS	Response or atlas out of the completed response scope.
TargetUpdate	Explicit richer-scope change requiring a new certificate.

A.1 Status distinctions

Status	Meaning
Complete	All target-authorized terminal uses are defined and positive on the fixed child scope.
Partial	The object carries some terminal work but lacks at least one mandatory use.
Ineligible	The object is not typed as an independent occupant of the terminal role.
Failed	An applicable terminal residual is negative.
Undefined	A required use is not formed in the declared source/target scope.
OneCarrier	Exactly one live bearer class and one assignment orbit within a realization.
OverResidue	More live cross-realization carrier classes than target roles.
CommonCore with target-dependent decoration	Cross-child carrier kind/core agrees, but exact target-relative standing packets differ.

B Vectorlike-joint source and sector normal form

B.1 Intrinsic source coefficients

For fixed $s = s_E > 0$, the source coefficients are

$$Y_{\nu}(m_0) = \frac{\sqrt{2}}{v} c_E U \Sigma(m_0) V^{\dagger}, \quad Y_{LN}(\lambda) = -\frac{\sqrt{2}}{v} s_E H_{\lambda} U,$$

$$\mu(m_0) = s_E \Sigma(m_0) V^{\dagger}, \quad M_N(\lambda) = c_E H_{\lambda} \mathbf{1}_3.$$

The source operators are

$$-\bar{L}\tilde{H}Y_{\nu} \nu_R - \bar{L}\tilde{H}Y_{LN} N_R - \bar{N}_L \mu \nu_R - \bar{N}_L M_N N_R + \text{h.c.}$$

The exact factorization is

$$\mathbb{M} = \mathcal{U}_L(s_E) \text{diag}(\Sigma(m_0), H_{\lambda} \mathbf{1}_3) \mathcal{U}_R(s_E)^{\dagger}.$$

B.2 Carrier-bearing sector normal form

Sector	Support	Standing witness	Lawful quotient
S_1	First simple light spectral root	Relative phase, boundary incidence, D1/D4 light response	Rephasing/permutation covariance
S_2	Second simple light spectral root	Independent relative phase and boundary incidence	Rephasing/permutation covariance
S_3	Third simple light spectral root	Independent relative phase and boundary incidence	Rephasing/permutation covariance
S_H	Triply degenerate heavy subspace	D3 threshold, active closure, D5 matching	Internal $U(3)$ degenerate-subspace action

B.3 Relation decoration

The relation decoration is generated by:

$$B_{CC}, \quad R_{LH}, \quad R_U, \quad R_P, \quad R_G^b, \quad R_{\beta\Sigma}, \quad R_V(z), \quad R_\Lambda.$$

The exact-global branch omits R_V as an active source and includes the exact protector in $R_G^{b_{\text{glob}}}$. The accidental branch permits $R_V(z)$ on the nonzero source stratum and uses the bounded accidental action $R_G^{b_{\text{acc}}}$.

B.4 No-repair deletion matrix

Deleted object	Disposition	First target loss
Any light sector	Fail	Exact oscillation/endpoint/population packet changes without retuning.
All light sectors	Fail	Oscillation, D1, D4, and light boundary disappear.
Heavy sector	Fail	D3 heavy-state/active-closure and D5 threshold matching disappear.
Light-heavy incidence	Fail/Undefined	Source-connected mass map and matching continuation break.
D4 population relation	Fail	Public-summary population/background target absent.
D5 child action/protector	Fail	Branch-specific D5 target absent.
Accidental LNV relation	Survives at zero subfibre	D2 response changes but target remains accepted and carrier core unchanged.
Common-cone relation	Fail	One-apex compatibility lost.
Heavy basis label	Skin	Complete trace unchanged under lawful $U(3)$.

C Terminal signature classification

The full terminal signature is evaluated over the Cartesian product of the activated witness family and continuation category. The following table records the decisive support-level result; the supplementary data archive contains the expanded object-by-witness representation.

Object	Def	Bdy	Pers	Cont	Comp	Fail	Kind	Terminal status
S_1	P	P	P	P	F	F	P	Partial
S_2	P	P	P	P	F	F	P	Partial

Object	Def	Bdy	Pers	Cont	Comp	Fail	Kind	Terminal status
S_3	P	P	P	P	F	F	P	Partial
Proper light aggregate	P	P	P	P	F	F	P	Partial/Fail
S_L	P	P	P	P	F	F	P	Partial
S_H	P	P	P	P	F	F	P	Partial
Proper light-heavy subset	P	P	P	P	F	F	P	Partial/Fail
Population relation	P	F	F	P	P	P	F	Ineligible
Symmetry / protector	P	F	F	P	P	P	F	Ineligible
LVN relation	P	F	F	P	P	P	F	Ineligible
Matching / higher relation	P	F	F	P	P	P	F	Ineligible
Coordinate-total parent-different candidate	P	P	P	P	P	P	P	Different/Ineligible
Parent carrier core	P	P	P	P	P	P	P	Complete

Here P means the corresponding block is positively defined and F means that the object lacks a required independent terminal use in that block. A row may be coordinate-total and parent-different without being ruled out by eligibility alone. P6-S2A forms eligibility independently; P6-S2B proves that an eligible, terminal-role-complete, one-apex compatible candidate inside the fixed realization must contain the full parent support and therefore cannot remain a distinct parent-different bearer. The table is not a substitute for the full continuation signature; it summarizes the missing-work and live-bearer classification proved in the main text.

C.1 Assignment orbit

For every ready X :

$$\begin{aligned}
 R_{\text{term}}^{\nu,b} &= \{r_{\text{bear}}^b\}, \\
 \text{Residue}_{\text{term}}^{\text{live}}(X) &= \{[C_{\nu}^{\text{VJ}}(X)]\}, \\
 \Sigma_{C_X}(T_{\text{term}}^{\nu,b}) / \Gamma_{C_X,T}^{\text{lawful}} &= \{[\sigma_X]\}.
 \end{aligned}$$

D Sources and dependencies

ID	Manuscript	Role in PCR-6
P6-S01	<i>A Fixed-Domain Exhaustion Theorem</i>	Same-scope exhaustion, fail-closed typing, anchor/tensor/skin closure.
P6-S02	<i>Role-Occupancy Closure under Admissibility</i>	Eligibility, compatibility, lawful automorphisms, unique assignment orbit.
P6-S03	<i>Standing Carrier Identity Closure</i>	Complete continuation signatures, activation, unique same-scope identity quotient.
P6-S04	<i>Constraint-Induced Physical Carrier Realization ARC Plan</i>	PCR-6 remit and endpoint discipline.
P6-S05	<i>Target-Fixed Physical Role Necessity for the Realized Neutrino Oscillation Interface</i>	Carrier-neutral role and nonalignment/response requirements.
P6-S06	<i>Complete Physical Interface of the Neutrino Propagation Role Complex</i>	Boundary reconstruction and evaluator completeness.
P6-S07	<i>Source-First Exhaustion of the Physical Carrier Universe</i>	Source-first sector reconstruction and activity reduction.

ID	Manuscript	Role in PCR-6
P6-S08	<i>Simultaneous Physical Constraint Collapse of the Neutrino Carrier Universe</i>	One-apex physicality and source-kind boundaries.
P6-S09	<i>Compatible Role-Occupancy Closure for the Neutrino Propagation Role Complex</i>	Internal factorization versus terminal carrier projection.
P6-S10	<i>Standing-Carrier Upper Closure for the Neutrino Propagation Role Complex</i>	Seven-channel packet and source-deletion standing.
P6-S11	<i>Exact Standing-Carrier Stratification of the Neutrino Oscillation Realization Space</i>	Exact base atlas, B15 interface, threshold/matching standing.
P6-S12	<i>Functorial Target Extension of Standing-Carrier Quotients</i>	Fibre and joint-target calculus.
P6-S13–18	PCR-5D1 through PCR-5D5	Marginal standing coordinates, target tags, and failure boundaries.
P6-S19	<i>Carrier Identity versus Spectral, Conjugation, Closure, and Transport Standing</i>	Cardinality separation and the PCR-6 identity problem.
P6-S20–21	Mini-ARC theorem plan and refinement addendum	PCR-5E/PCR-6 architecture and coordinate diagnostics.
P6-S22	<i>Joint Standing-Carrier Closure for the Neutrino Sector</i> , v1.4.0	Exact completed child sectors and E-H1–E-H10 imports.
P6-S23–24	MGN5 and MGN6	Matrix/spectrum/mixing no-promotion.
P6-S25	<i>The Neutrino-Mass Low-Energy Exact-Character Endpoint</i>	Mass-role and operator-availability discipline.
P6-S26–28	Higgs Structural Endpoint Arc	Higgs identity, scale, extension, and projection no-backflow.
P6-S29–31	Standard Model carrier decomposition arc	Matter-carrier, flavor, generation, and primitive inventory distinctions.
P6-S32	PCR-5E D3 Successor-Branch ARC Target-Update Memorandum	D3-1.3 branch authority and D3-1.1 preservation.
P6-S33	FCGS Mini-ARC Detailed Skeleton	Downstream generation-sector and universality boundary.

E Theorem dependencies

ID	Theorem	Principal dependencies
P6-00	Import and no-backflow lock	PCR-5E v1.4, ARC-U1, fixed-domain, SCIC.
P6-R1	Ready scope and parent nonpromotion	E-H data and parent readiness boundary.
P6-01	Branch-indexed finality	E-H1–E-H10.
P6-02	Physical-nature targets	PCR-2, PCR-5E, ROC.
P6-03	Semantic sector class	PCR-2 semantic uses, SCIC extensionality.
P6-S0	Physical-nature semantic-use descent	PCR-3 carrier trace, PCR-5E E-S1/E-S2, fixed-domain factorization.
P6-04	Source-first sector grammar	P6-S0, PCR-3, PCR-5B, E-S1/E-S2.
P6-05	Activity reduction	PCR-3/PCR-5E source activity.
P6-S1	Complete heavy-degenerate witness descent	P6-S0, VJ mass map, D3/D5 witness blocks, lawful $U(3)_H$.
P6-06	Exact 3 + 1 spectral core	P6-S1, VJ mass map, MGN5, D5 degeneracy/action.
P6-07	Cardinality neutrality	Source ancestry and no-promotion.
P6-R2	Moduli-natural sector functor	P6-S0, P6-05–06, PCR-5C groupoids, E-H5.

ID	Theorem	Principal dependencies
P6-08	Boundary incidence	PCR-2, MGN6, VJ boundary blocks.
P6-09	Continuation category	E-H7, PCR-5C composition.
P6-10	Terminal activation	P6-S0, fixed-domain/SCIC activation plus E-H fields.
P6-11	Sector signatures	P6-06, P6-08–10, deletion packets.
P6-S2A	Target-derived terminal eligibility	P6-02, P6-S0, P6-08–10, general ROC formation rules.
P6-S2B	Same-apex complete-bearer exhaustion and parent-support uniqueness	P6-S0, P6-04, P6-11–12, deletion matrix, one-apex physicality.
P6-12	Parent signature	PCR-5E one-apex physicality, semantic saturation, P6-10–11.
P6-13	Promotion necessity	P6-S2A/S2B, SCIC and ROC.
P6-14	Spectral nonpromotion	P6-11, P6-S1, P6-S2A/S2B, MGN5/6, Higgs arc.
P6-15	Positive one-carrier realization	P6-12, source-connected VJ core.
P6-16	Higher-arity alternative	Relation signatures and fixed-domain support exhaustion.
P6-17	Carrier-node exhaustion	P6-11–16 and P6-S2B.
P6-18	Carrier kind	VJ source normal form and PCR-3 kind classes.
P6-19	Scoped substitute finality	P6-S2A/S2B, E-H2 comparison category and SCIC.
P6-20	Locality/spin-statistics	PCR-2 physical interface and VJ local source.
P6-21	Quantum-number packet	D2/D5 and VJ anomaly/charge structure.
P6-R3	Coordinate inheritance and the τ -direction separation	PCR-5E E-S6/E-S7.
P6-22	Coordinate roles	P6-R3 and deletion/continuation analysis.
P6-23	Within-realization ROC	P6-S2A/S2B, P6-10–17, and general ROC theorem.
P6-24	Minimal identity	P6-15, P6-22, P6-R2.
P6-R2b	Moduli-natural nature packet	P6-R2, P6-S2B, P6-23–24.
P6-R4	Cross-child comparison	P6-24, PCR-5E D2/D5 relation.
P6-R5	Branch stability	P6-23, P6-24, P6-R4.
P6-25	Projection no-backflow	MGN5/6 and Higgs arc.
P6-26	Subsequent consequences	Entire completed chain.

F Supplementary formal records and reproducibility

The accompanying supplementary archive provides formal representations of the target scopes, source and sector structures, terminal-signature analysis, bearer-exhaustion calculation, physical-nature classification, Role-Occupancy results, coordinate relations, projection boundary, and unresolved parent scope. These records support navigation, provenance, and reproducibility; the scientific authority of the paper remains in the definitions and proofs of the main text.

Record class	Scientific content
Scope and target data	Target and comparison fingerprints for the two completed vectorlike-joint sectors, branch-indexed physical-nature targets, activated witnesses and continuations, and the unresolved parent boundary.
Semantic-use and source data	Relative completeness of the physical-nature use trace over the PCR-3/PCR-5E source, source-first sector normal forms, activity reduction, lawful overlaps, and typed complements.
Spectral-sector data	The four-sector carrier-bearing core, lawful relation decoration, moduli strata, and channel-by-channel descent of the heavy witness family through the $U(3)_H$ quotient.

Record class	Scientific content
Terminal-signature data	Complete, partial, different, ineligible, undefined, and update-bearing dispositions for sector nodes, aggregates, irreducible relations, the source-connected parent, and parent-distinct candidates.
Eligibility and bearer exhaustion	Target-derived terminal eligibility formed independently of SCIC identity, together with same-apex source-support exhaustion and reduction of every complete compatible candidate to the parent carrier class.
Carrier and physical-nature data	Carrier-node separation, physical-kind classification, common carrier core, target-dependent decoration, minimal identity packets, and variation across the solved moduli.
Role-Occupancy data	The PCR-5E cross-realization over-residue calculation and the independent PCR-6 within-realization singleton assignment orbit.
Coordinate and projection data	Common generation, obstruction, independent-coordinate relations, the fixed-base τ -direction, premetric-to-metric projection, Higgs/Yukawa and pole-status interfaces, and mixing-boundary discipline.
Scope and downstream data	The unresolved parent sectors, prohibited extensions of the child theorem, and target-qualified consequences for subsequent observable and fermion-sector analyses.

The supplementary consistency checks verify target tags, theorem references, multiplicity distinctions, heavy-witness descent, independent eligibility, same-apex support uniqueness, assignment-orbit counts, parent-scope boundaries, and file integrity. They do not replace the semantic proofs of source completeness, terminal activation, one-apex physicality, or Role-Occupancy Closure.

G Deletion and continuation witness matrix

This appendix records the concrete witness changes used in the bearer-minimality and sector-nonpromotion proofs. It is a compressed representation of the complete continuation signatures, not a replacement for them.

G.1 Carrier-support deletions

Deleted support	First changed witness	Consequence
S_i for one light projector	Retained phase-frequency and boundary-incidence packet	The exact oscillation response loses one nondegenerate propagation component; D1 endpoint and D4 light-population weights no longer agree with the accepted child.
Any two-light proper aggregate	No-repair oscillation and endpoint reconstruction	The remaining light support cannot reconstruct the three-root relative spectrum or the complete electron-boundary spectral measure.
Full light aggregate S_L	D3 heavy-state and D5 threshold/matching witness	The child loses the active heavy response, heavy threshold, and oriented matching data.
Heavy sector S_H	D3 H/U response and D5 vectorlike matching	The accepted active-closure and symmetry/matching packet is destroyed.
Light-heavy source incidence	Source connectivity and exact mass-map factorization	The intrinsic source splits, the heavy charged-current block is no longer derived from the common source, and the matching continuation is undefined.
Charged-current boundary incidence	Boundary reconstruction	Neither light oscillation nor heavy production/detection has the accepted operational boundary.
D4 population/background relation	Public-summary target record	The accepted standard-thermal stress-energy and mass-sum response is not realized.
Exact-global protector	Exact-global continuation and D2 zero subfibre	The object leaves the exact-global child target.
Accidental LNV relation at $z \neq 0$	D2 response amplitude	The enriched accidental class changes, but the $z = 0$ child point and common carrier core remain lawful.
Common-cone relation	One-apex compatibility	Marginal witnesses cease to be certified as one physical realization.

G.2 Continuation tests

Continuation	Invariant support	Identity test
Oscillation propagation	Three light sectors and boundary map	Relative phase and boundary reconstruction are preserved; deletion of a light sector is detectable.
D3 heavy propagation/decay	Heavy sector, active block, threshold	The heavy rank-three aggregate is retained; individual heavy basis labels are gauge/basis representatives.
Finite-order RG/EFT matching	Light-heavy relation and oriented threshold	All retained light outputs descend while the heavy threshold remains standing-visible.
D1/D4 τ continuation	Same intrinsic source and all seven PCR-5B base coordinates	D1/D4 enriched response changes without changing the terminal bearer partition.
D4 kinetic/background continuation	Three active light population slots and common mass record	Source-false vectorlike-heavy D4 incidences remain null; the intrinsic D3/D5 heavy source is not deleted.
D5 accidental continuation	Common carrier core plus bounded $d \leq 4$ action	Higher-dimensional LNV may vary as decoration.
D5 exact-global continuation	Common carrier core plus protector	D2 remains at the defined zero subfibre and exact charge action is continuation-stable.
Lawful source/basis redescription	Complete source-connected carrier trace	Presentation, heavy basis, and matrix-coordinate changes do not create new sectors or bearers.

Proposition G.1 (Deletion/continuation completeness of the minimal core). *Every field of the common carrier core appears in at least one row of the preceding deletion matrix, and every licensed continuation in the ready child acts on that core or its target-standing decoration. Hence no retained common-core field is witness-free, and no omitted presentation field is required for identity-preserving reuse.*

Proof. The carrier-support rows cover light support, heavy support, common boundary, source connectivity, and matching. The continuation rows cover propagation, RG/EFT, absolute-scale, population/background, symmetry, and lawful redescription. These are precisely the terminal signature blocks activated in the main text. Matrix coordinates, field names, and heavy basis vectors do not appear because their changes factor through lawful presentation automorphisms. $\square$

Proposition G.2 (Full-support subobject inheritance). *Let $A \hookrightarrow X$ be a source-faithful same-apex subobject in a ready E-H2 child. If A contains every deletion-essential support and relation of the minimal carrier core, then the source parameters, target attachments, boundary data, matching orientation, continuation laws, and active incidence values of A are the restrictions of those of X . Any remaining difference is lawful presentation/skin or target-inert enrichment. Hence*

$$\Sigma_{\text{term}}^b(A) = \Sigma_{\text{term}}^b(X)$$

after lawful reduction. A full-support object with a different standing-bearing value is not a subobject of the fixed realization; it belongs to a different joint realization class or to a declared target/source update.

Proof. The within-realization source category is a restriction category over the accepted E-H2 source point. Its morphisms retain or delete source support and incidence while preserving the inherited source equations and parameter values; they do not retune them. Once every deletion-essential support is retained, the carrier-facing PCR-3 trace and simultaneous PCR-5E incidence trace therefore agree with those of X on every terminal-bearing coordinate. Lawful source, basis, gauge, chart, and heavy-degenerate rotations preserve those traces, while target-inert extra presentation fields are skin. Changing a standing-bearing source or continuation value instead moves to another source point and therefore another PCR-5E realization or an explicit update. Terminal activation then gives equality of the complete continuation signatures for every genuine full-support subobject of the fixed realization. $\square$

H Claim boundaries and counterexample scope

H.1 Established claims

Claim	Exact authority
Completed-sector scope	The theorem applies to the two PCR-5E vectorlike-joint child comparison classes only.
Internal sector quotient	Every accepted child realization has three light standing spectral sectors, one rank-three heavy standing sector, and a target-indexed relation decoration.
Terminal bearer cardinality	Every accepted child realization has one live terminal bearer class and one lawful assignment orbit.
Physical kind	The terminal bearer is a local hybrid neutral-fermion carrier at the intrinsic source level, with vectorlike Dirac mass-map core and chiral charged-current boundary.
Moduli naturality	Bearer cardinality and $3 + 1$ sector topology are preserved under lawful transport across each completed sector groupoid.
Cross-child relation	The common carrier core agrees; exact-global and accidental symmetry/conjugation decorations remain target-distinct.
Projection boundary	Metric, Higgs/Yukawa, mass-matrix, pole, and mixing presentations cannot generate a new premetric terminal partition.

H.2 Claims not made

The paper does not claim any of the following:

1. that the PCR-5E child quotient is singleton or that one joint realization is selected;
2. that the unresolved PCR-5E parent source universe has the same physical nature;
3. that D3-1.1 or the complete D4 cosmological target is completed;
4. that the four standing sectors are four elementary particles or four terminal carriers;
5. that the exact heavy degeneracy persists under a richer target or source update;
6. that the accidental and exact-global complete identity packets are equal;
7. that every neutrino realization, charged lepton, or quark has one terminal carrier;
8. that numerical masses, Yukawa values, or PMNS entries are derived;
9. that a point-local QFT presentation is the premetric identity ground rather than a faithful downstream realization;
10. that supplementary records or automated checks prove any semantic theorem.

H.3 Same-scope counterexample conditions

A same-scope counterexample must preserve the child target fingerprint, source-qualified E-H2 comparison category, one-apex physicality, terminal witness/continuation family, lawful equivalence, and failure semantics. It must then exhibit one of the following: a physical-nature use not descending through the complete PCR-3/PCR-5E trace; an omitted active sector; an independently eligible, terminal-role-complete, one-apex compatible candidate not reduced to the parent by P6-S2B; a second compatible terminal bearer; a relation with independent persistence and boundary reconstruction; an activated witness separating heavy basis vectors despite exact degeneracy and lawful $U(3)_H$; or a downstream projection that changes terminal identity without an antecedent premetric standing difference. A different source architecture in the unresolved parent complement defines a legitimate extension of the comparison problem, but it is not a counterexample to the theorem at the completed child scope.

I Dependency structure and downstream scope

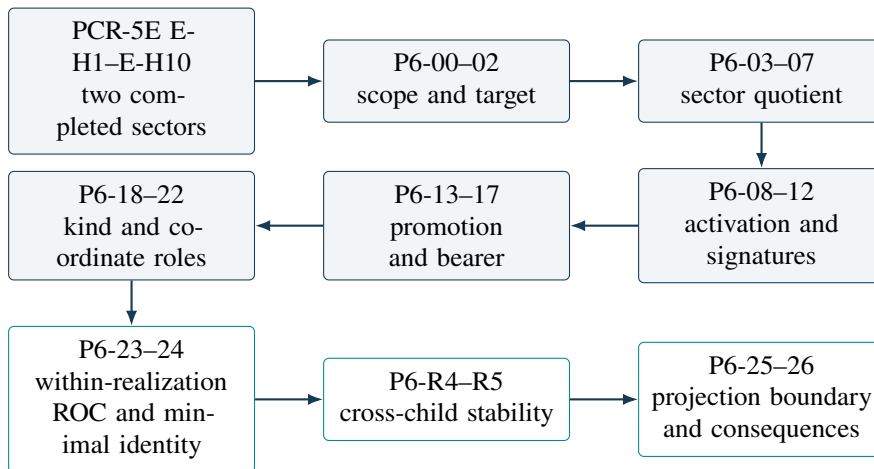


Figure 3: Dependency structure of the PCR-6 theorem. The endpoint follows only after sector completeness, terminal activation, complete signature comparison, physical parent realization, and the within-realization Role-Occupancy calculation.

Result	Object	Scope of subsequent use
PCR6-H1	Completed-sector target tags and unresolved parent boundary	No transport to unresolved parent branches.
PCR6-H2	Natural sector functor and exact $3 + 1$ core	No identification with generation roles outside the child source category.
PCR6-H3	One terminal bearer and assignment-orbit certificate	Does not imply singleton PCR-5E quotient.
PCR6-H4	Common carrier core and child decorations	Accidental/exact-global complete packets remain distinct.
PCR6-H5	Projection boundary	Metric/Yukawa displays cannot add premetric identity.

Result	Object	Scope of subsequent use
PCR6-H6	Failure and update frontier	New source kinds or lifted heavy degeneracy require explicit target/source updates.

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